

Electrical Conduction in Solids

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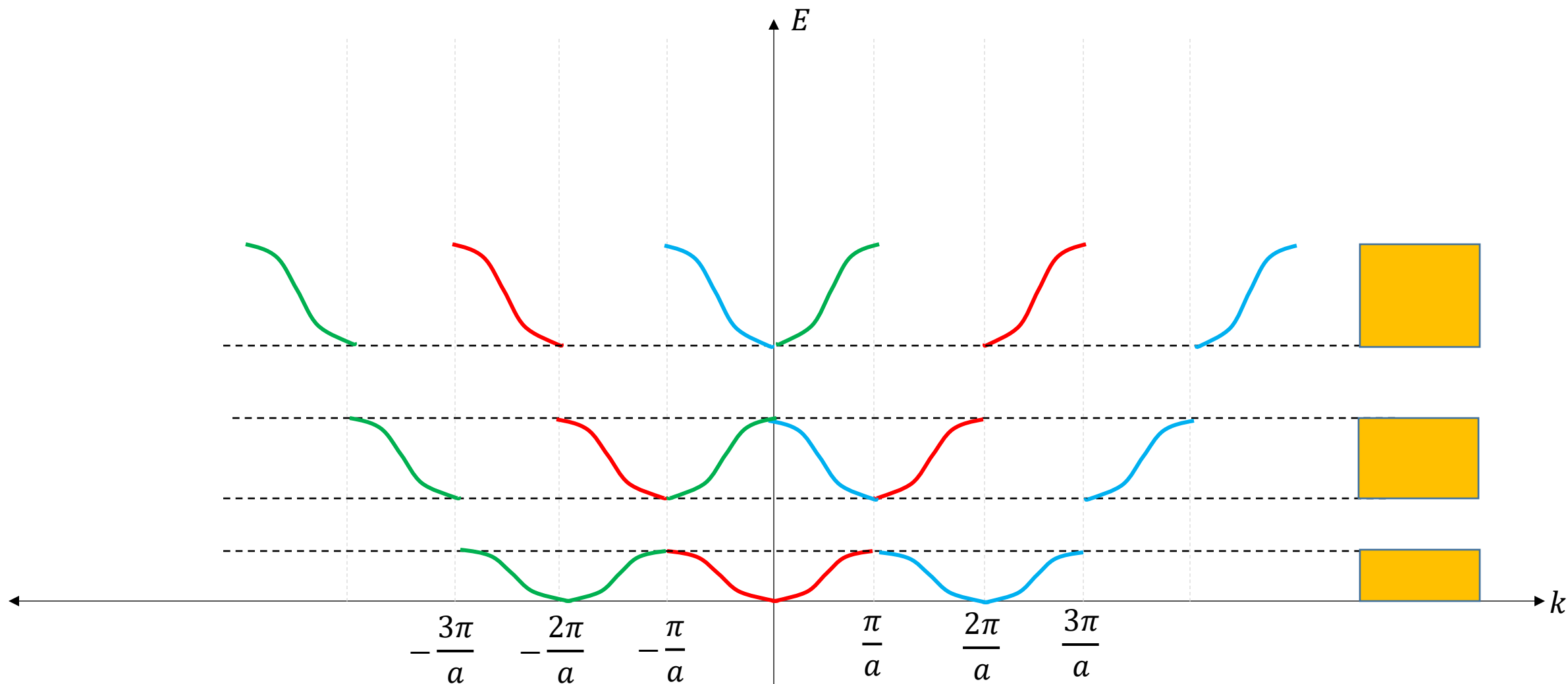
K-P model

$$\cos ka = 1$$
$$ka = \pm n\pi$$

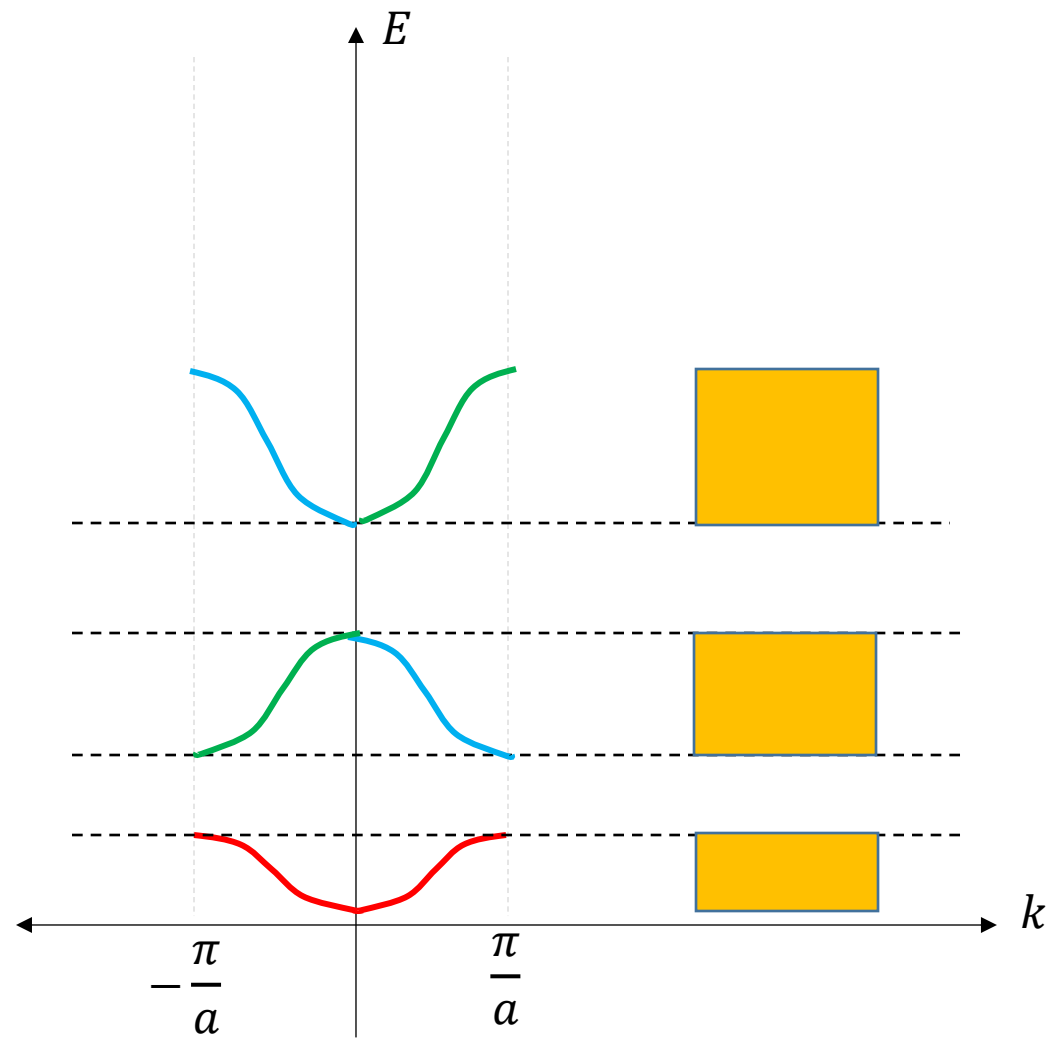
$$k = \pm \frac{n\pi}{a}$$

$$k = \pm \frac{\pi}{a}, \pm \frac{2\pi}{a}, \pm \frac{3\pi}{a} \dots \dots$$

$$E = \frac{k^2 \hbar^2}{2m}$$



Reduced k -space



The BOND Model

Consider Silicon

- Each Si atom is surrounded by 8 valence electrons
- All electrons shown in Fig. 1 are valence electrons
- At $T = 0K$, all the electrons are present in the valence band
- At $T = 0K$, no free electron present, so the conduction band is completely empty.

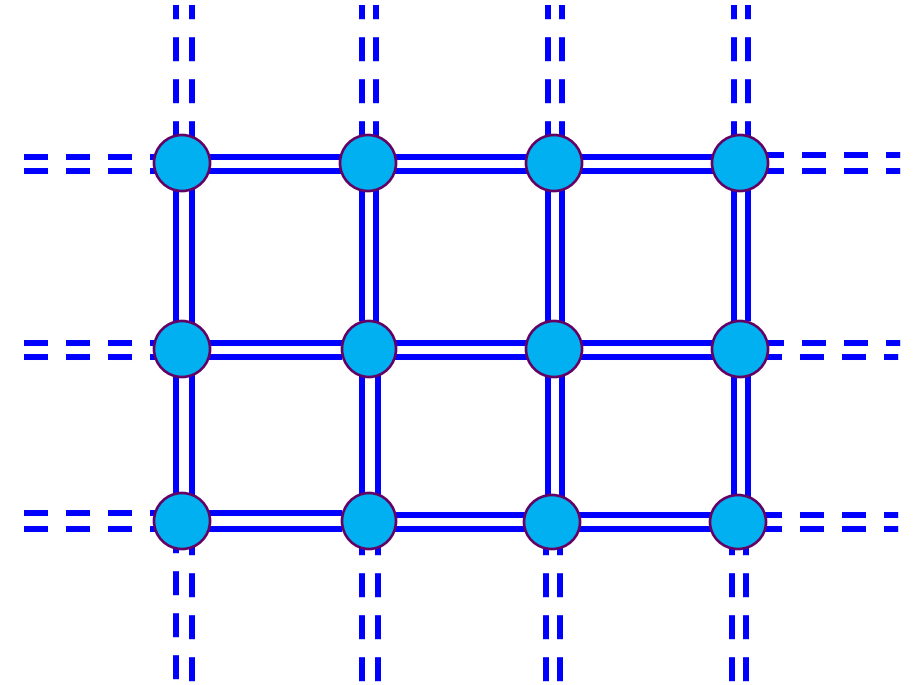
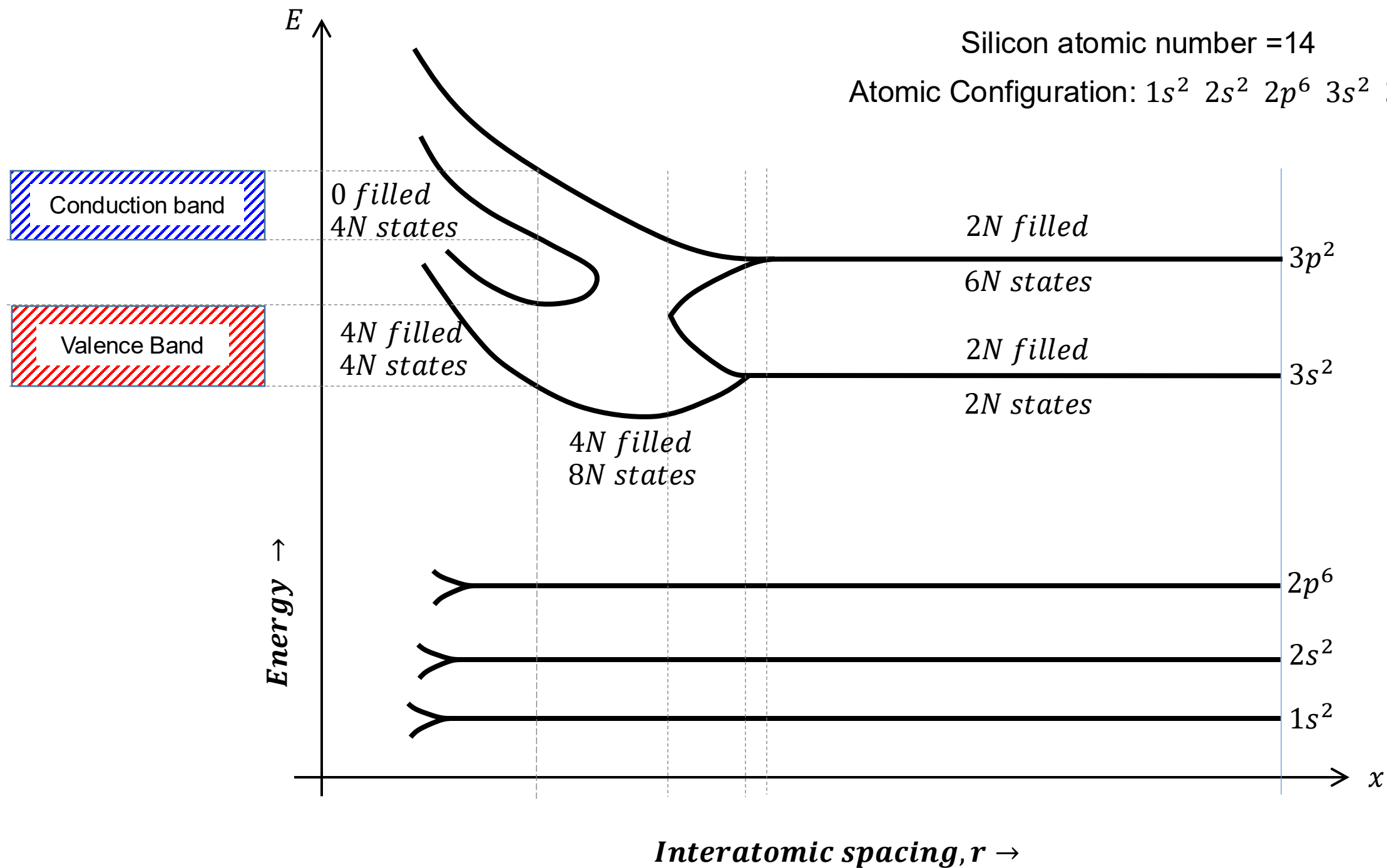


Fig. 1: Two dimensional representation of covalent bonding in a semiconductor at $T = 0K$

Silicon atomic number = 14

Atomic Configuration: $1s^2 2s^2 2p^6 3s^2 3p^2$



The BOND Model

- At $T > 0K$, few valence electron may gain enough thermal energy to break covalent bond and jump into conduction band
- The semiconductor is neutral. As the negatively charged electron breaks away from its covalent bonding position, a positively charged “empty state” is created in the original covalent bonding position in the valence band.
- If we further increase the temperature, more covalent bonds will be broken generating more electrons in the conduction band and more empty states in the valence band.

- **Thermal generation**
- **Carrier generation in pairs due to breaking of covalent bond**

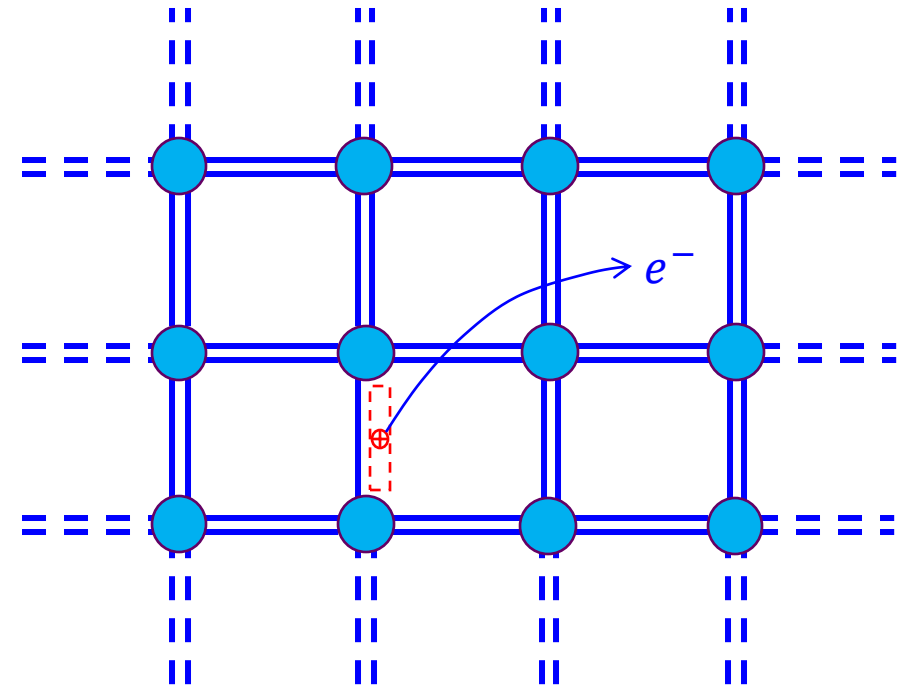


Fig. 2: Two dimensional representation of covalent bonding in a semiconductor at $T > 0K$

The BOND Model

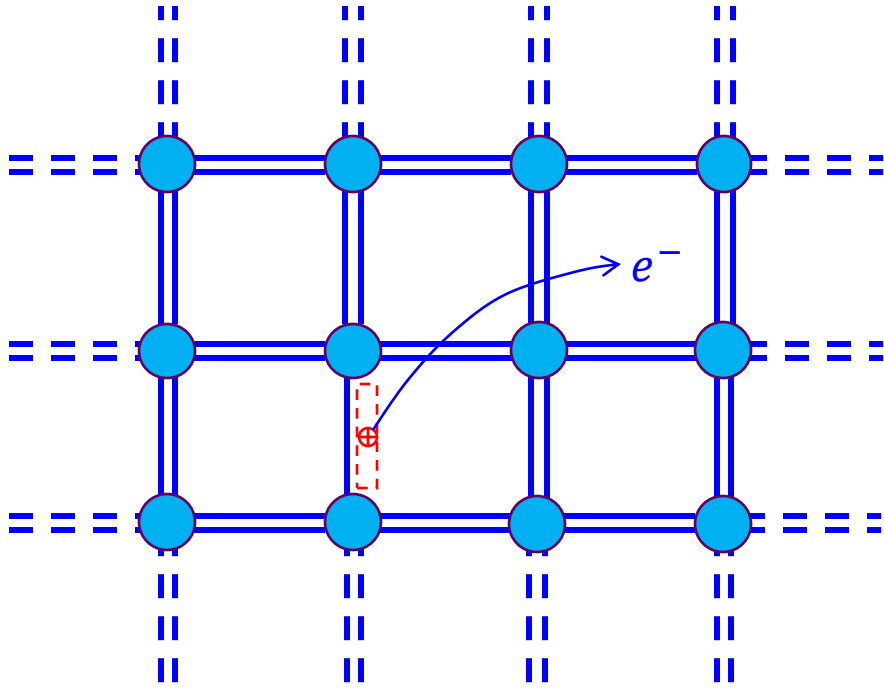


Fig. 2: Two dimensional representation of covalent bonding in a semiconductor at $T > 0K$

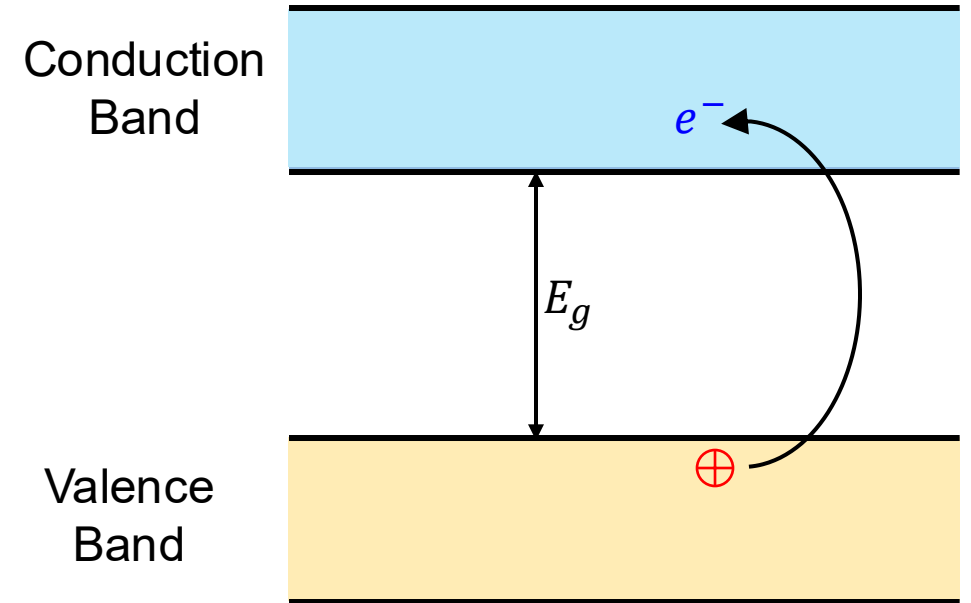


Fig. 3: Energy band representation and generation of electron and hole in a semiconductor at $T > 0K$

The BOND Model

- Electrons that break away from covalent bond (free electron)
- ⊕ Positively charged empty state

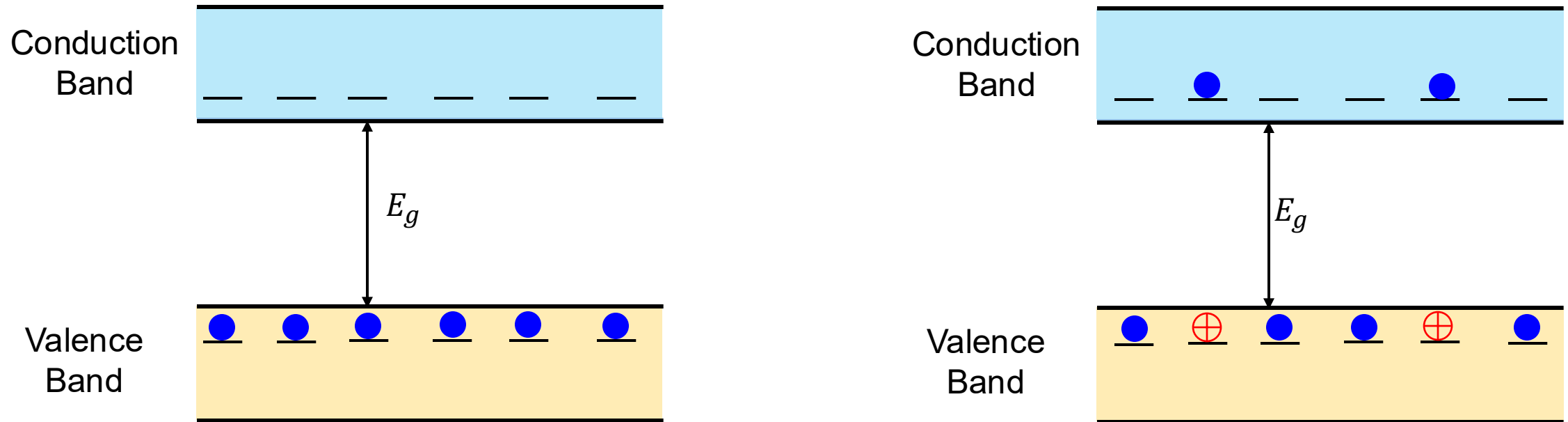


Fig. 4: Energy band representation and generation of electron and hole in a semiconductor at
(a) $T = 0K$ and (b) $T > 0K$

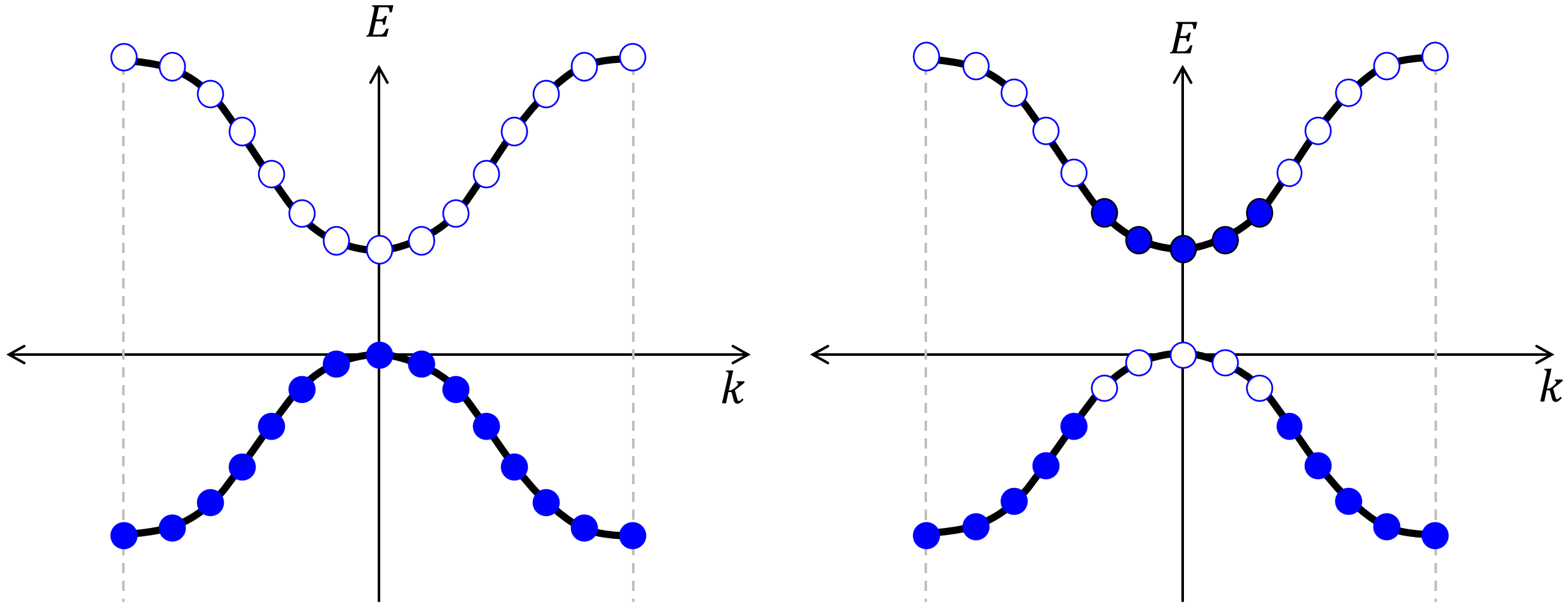


Fig. 5: E vs k diagram of the valence band and conduction band of a semiconductor at
(a) $T = 0K$ and (b) $T > 0K$

If no external forces are applied, electron and the empty state distribution will be symmetrical with k

Drift current

Let us consider a collection of positively charged ions with a volume density N (cm^{-3}) and an average drift velocity v_d (cm/s).

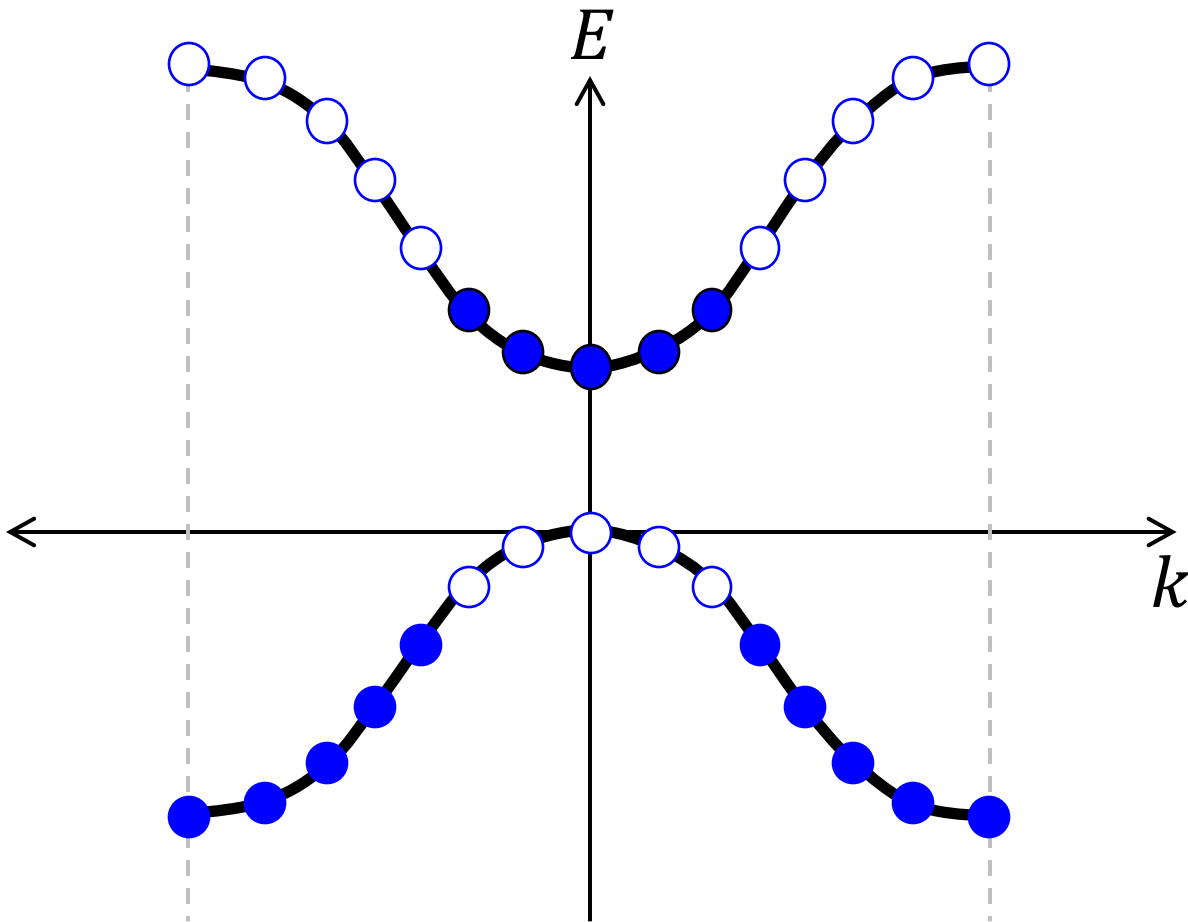
Then the drift current density can be written as,

$$J = qNv_d \quad A/cm^2$$

Instead of considering the average drift velocity, if we consider individual ion velocities, then

$$J = q \sum_{i=1}^N v_i$$

Since electrons are charged particles, a net drift of electrons in the conduction band will give rise to a current



- The electron distribution in the k – space is an even function of k
- k for a free electron is related to its momentum

$$\left(k = \frac{2\pi}{\lambda} = \frac{2\pi}{\frac{h}{p}} = \frac{p}{\hbar} \right)$$
- This means, there are as many electrons with a $+|k|$ value as there are with a $-|k|$, value.
- The net drift velocity due to these electrons are zero
- Certainly expected result since there is no externally applied force.

What if an external force F is applied?

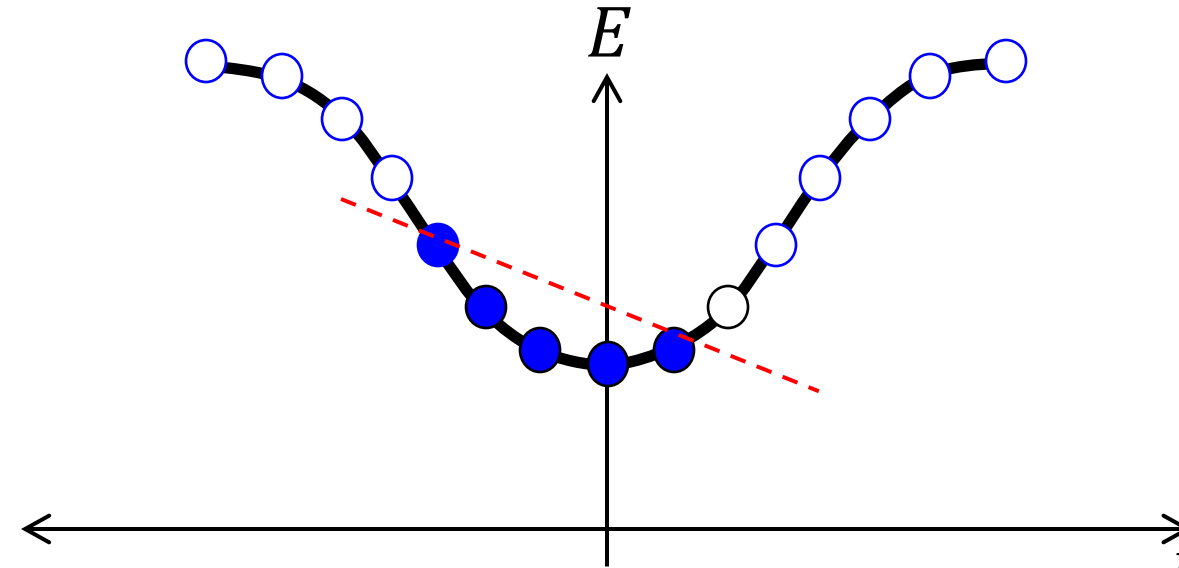
- Let us consider a force F is applied, the particle moves and it must then gain energy
- Let the particle moves with a velocity v
- If the particle moves a differential distance dx and dE is the increase in energy, then

$$dE = Fdx = F \frac{dx}{dt} dt = Fvdt$$

- If an external force is applied to the electrons in the conduction band, they gain energy and a net momentum as shown in Fig.
- The drift current density due to the movements of the electrons can be written as

$$J = -e \sum_{i=1}^N v_i$$

e is the electron charge
 v is the electron velocity
 N is the number of electrons



The current is directly related to electron velocity and current is related to how well the electrons can move in a crystal.

Concept of effective mass

We know, $F = ma$

If we consider that an electron in free space has mass m and an external force F_{ext} is applied on it then it will accelerate with acceleration ' a ' and we can write,

$$F_{ext} = ma \rightarrow (1)$$

The movement of an electron in lattice will in general be different from that of an electron in free space. This is primarily due to the fact that, in addition to F_{ext} , an electron in lattice will experience internal forces in the crystal due to the positively charged ions or protons and negatively charged electrons. If we assume the internal forces to be F_{int} , we can write total force F_{total} acting on the electron as

$$F_{total} = F_{ext} + F_{int} = ma \rightarrow (2)$$

Since it is difficult to take into account of all the internal forces, in general, equation (2) is re-written as,

$$F_{ext} = m^*a \rightarrow (3)$$

The parameter m^* is called the effective mass of electron (particle) and it takes into account the electron (particle) mass and the effect of internal forces.

Effective mass in E versus k curves

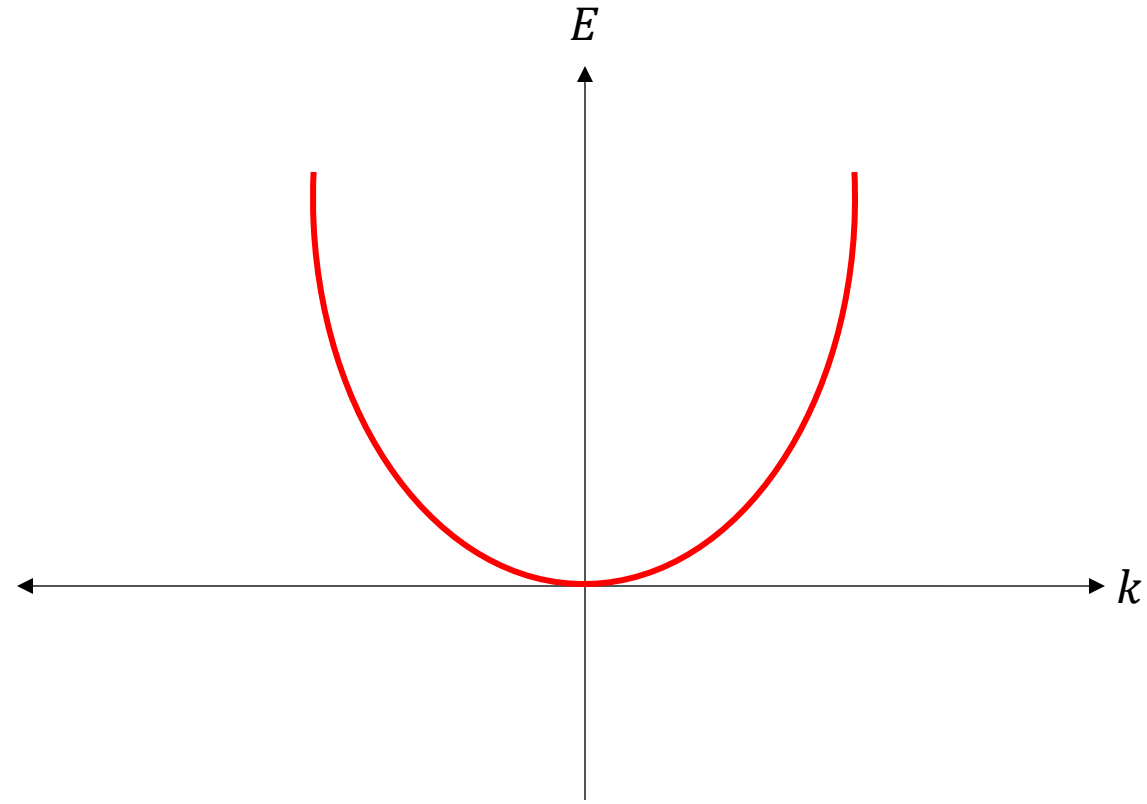
If we consider that an electron in free space,

$$E = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m} = Ck^2 \text{ (parabola)}$$

$$\frac{dE}{dk} = \frac{2\hbar^2 k}{2m} = \frac{\hbar^2 k}{m}$$

$$\frac{dE}{dk} = \frac{\hbar^2 k}{m}$$

$$\frac{d^2 E}{dk^2} = \frac{\hbar^2}{m}$$



Effective mass in E versus k curves

$$p = \frac{h}{\lambda} = \frac{h}{2\pi} \cdot \frac{2\pi}{\lambda} \Rightarrow p = \hbar k$$

$$p = mv \Rightarrow v = p/m$$

$$\frac{dE}{dk} = \frac{\hbar^2 k}{m}$$

$$\frac{1}{\hbar} \frac{dE}{dk} = \frac{\hbar k}{m}$$

$$\frac{1}{\hbar} \frac{dE}{dk} = \frac{p}{m} = v$$

$$v = \frac{1}{\hbar} \frac{dE}{dk}$$

The first derivative of E with respect to k is related to the velocity of the particle

$$\frac{d^2 E}{dk^2} = \frac{\hbar^2}{m}$$

$$\frac{1}{\hbar^2} \frac{d^2 E}{dk^2} = \frac{1}{m}$$

$$\frac{1}{m} = \frac{1}{\hbar^2} \frac{d^2 E}{dk^2}$$

The second derivative of E with respect to k is related to the mass of the particle (inversely proportional)

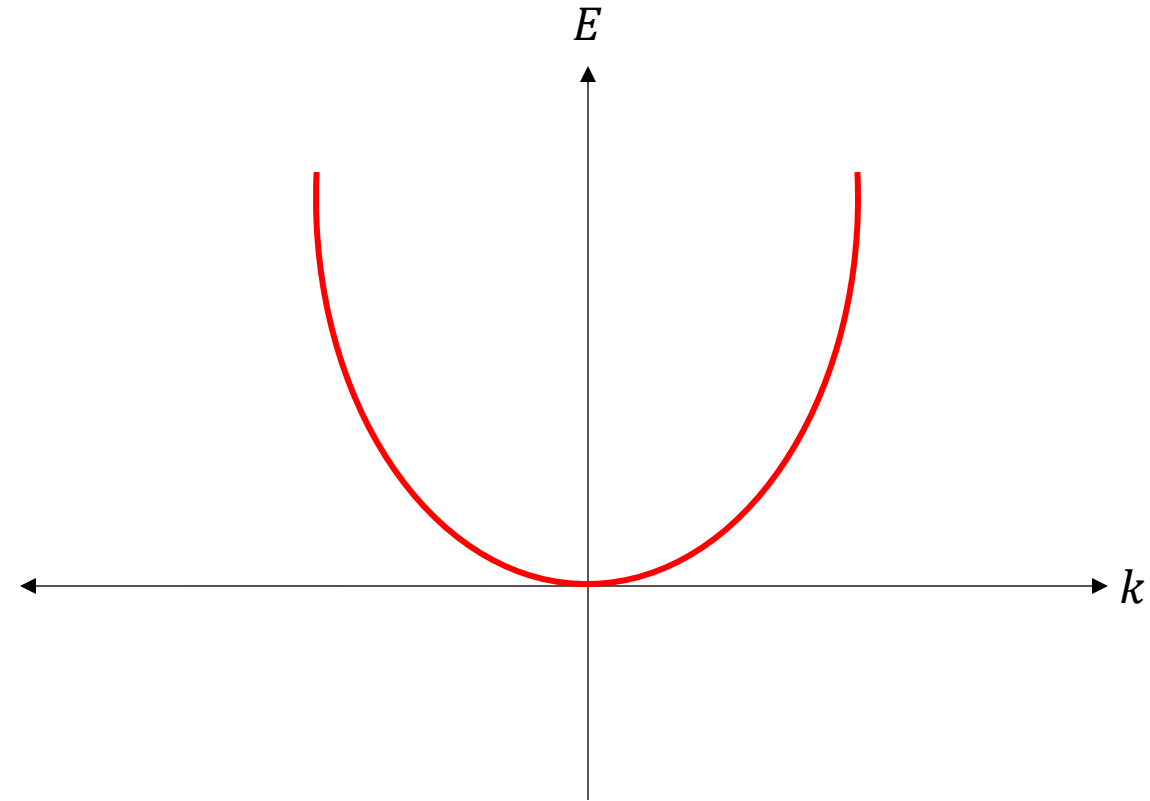
Effective mass in E versus k curves

If we consider that an electron in free space,

$$E = Ck^2 \text{ (parabola)}$$

$$\frac{d^2E}{dk^2} = 2C$$

$\frac{d^2E}{dk^2}$ is constant and positive



If we apply an electric field, E to the free electron then,

$$F = qE = -eE = ma$$

e is the electron charge
 a is the acceleration

$$a = -\frac{eE}{m}$$

The motion of electron is in the opposite direction to the applied electric field because of negative charge.

Effective mass at the bottom of the available/conduction band

The energy near the bottom of the available/conduction band may be approximated by a parabola,

$$E - E_c = C_1 k^2 \rightarrow (1)$$

$$E - E_c > 0 \Rightarrow C_1 > 0 \rightarrow (2)$$

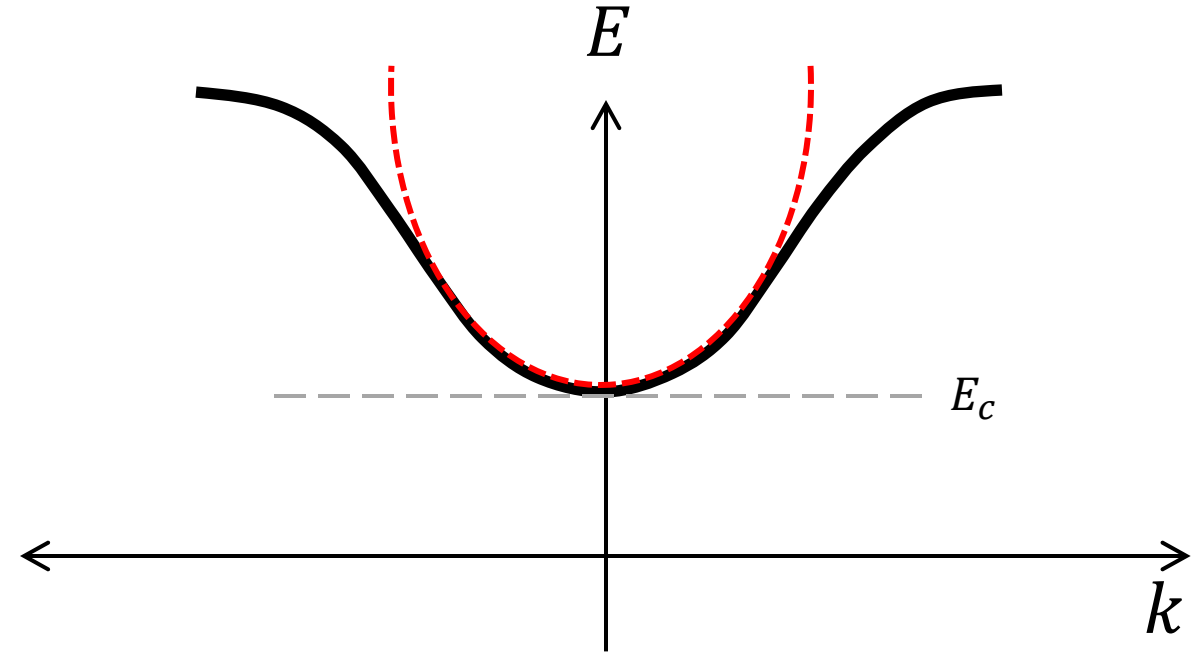
Taking 2nd derivative of E w.r.t k from equation (1), we get,

$$\frac{d^2 E}{dk^2} = 2C_1$$

$$\frac{1}{\hbar^2} \frac{d^2 E}{dk^2} = \frac{2C_1}{\hbar^2} = \frac{1}{m^*}$$

The parameter m^* is called the effective mass.

$$C_1 > 0 \Rightarrow m^* > 0$$



$$a = -\frac{eE}{m^*}$$

e is the electron charge
 a is the acceleration

Effective mass at the top of the available/valence band

The energy near the top of the available/valence band may be approximated by a parabola,

$$E_v - E = C_2 k^2$$

$$E_v - E > 0 \Rightarrow C_2 > 0 \rightarrow (1)$$

$$E - E_v = -C_2 k^2 \rightarrow (2)$$

Taking 2nd derivative of E w.r.t k from equation (2), we get,

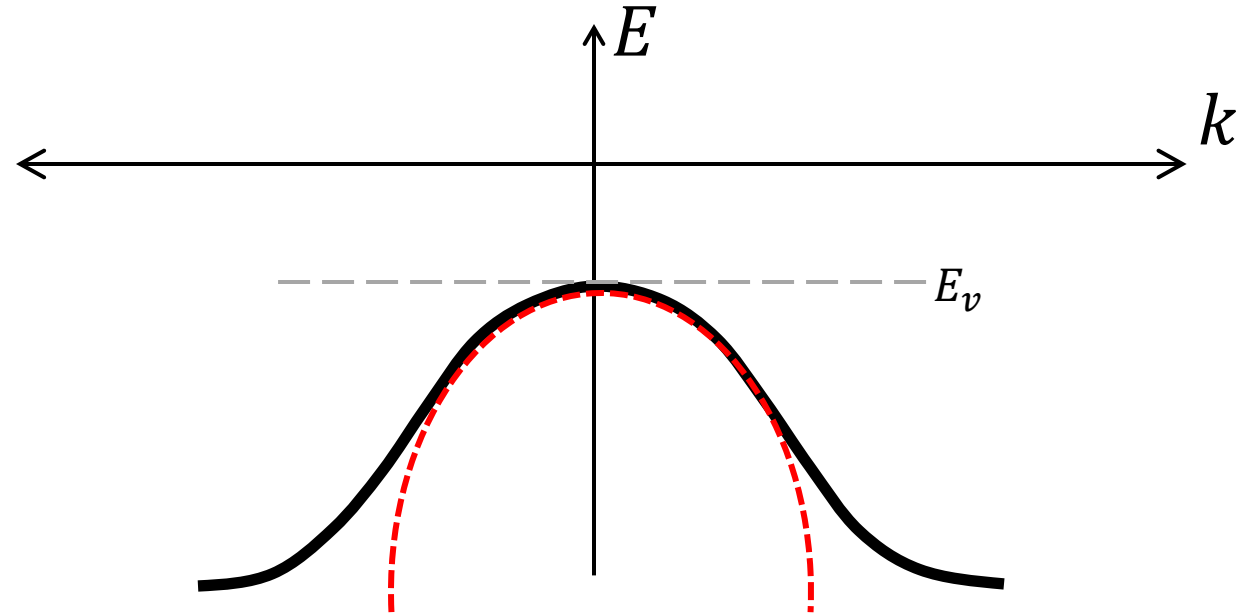
$$\frac{d^2 E}{dk^2} = -2C_2$$

$$\frac{1}{\hbar^2} \frac{d^2 E}{dk^2} = \frac{-2C_2}{\hbar^2} = \frac{1}{m^*}$$

The parameter m^* is called the effective mass.

$$C_2 > 0 \Rightarrow m^* < 0$$

An electron moving near the top of the valence band behaves as if it has a negative mass.



$$a = -\frac{eE}{m^*}$$

e is the electron charge
 a is the acceleration

$$a = -\frac{eE}{-|m^*|} = \frac{eE}{|m^*|}$$

An electron moving near the top of the valence band moves in the same direction of the applied field.

Concept of Hole

$$a = - \frac{eE}{-|m^*|}$$

Acceleration achieved by electron of charge $(-e)$ and mass $(-)$

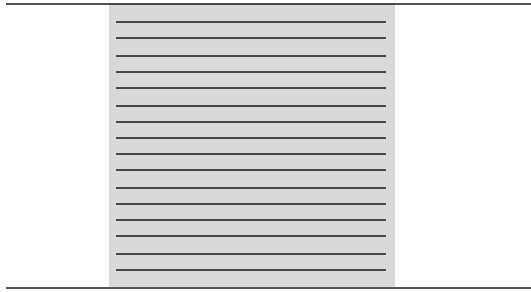
$$a = \frac{eE}{|m^*|}$$

Acceleration achieved by X of charge $(+e)$ and mass $(+)$

The net motion of electrons in a nearly full band can be described by considering just the empty states, provided that a positive electronic charge is associated with each state and that the negative of m^* is associated with each state.

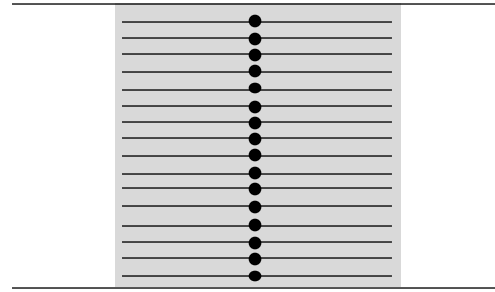
This new particle is called hole

Metals, Insulators and Semiconductors



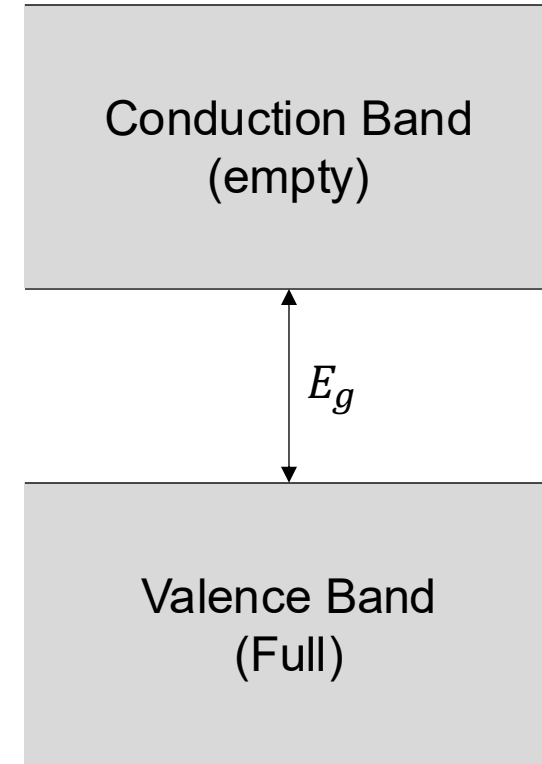
Allowed Energy Band
(empty)

- ✓ Apply electric Field
- ✓ No free electron
- ✓ No Current

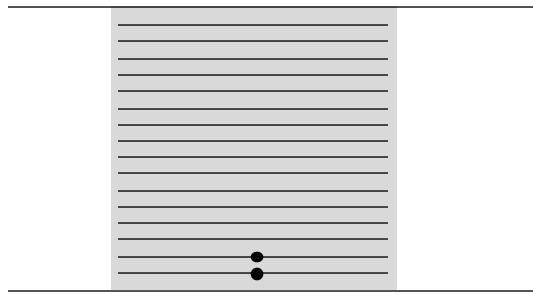


Allowed Energy Band
(Full)

- ✓ Apply electric Field
- ✓ Average drift velocity=0
- ✓ No Current

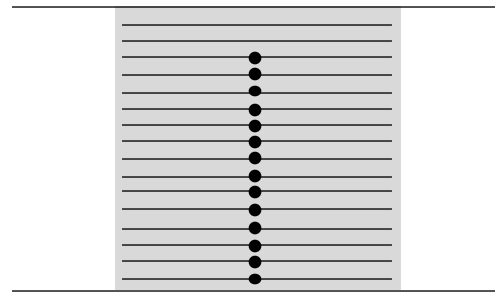


Metals, Insulators and Semiconductors



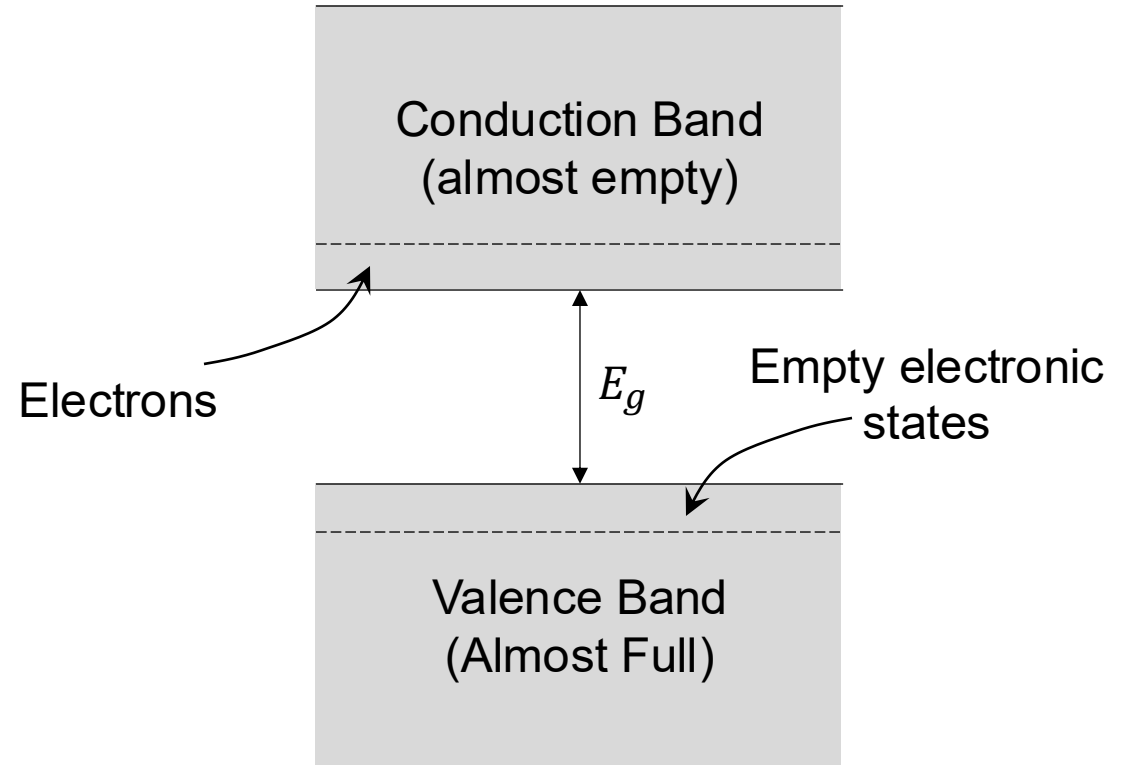
Allowed Energy Band
(empty)

- ✓ Apply electric Field
- ✓ Free electron
- ✓ Current

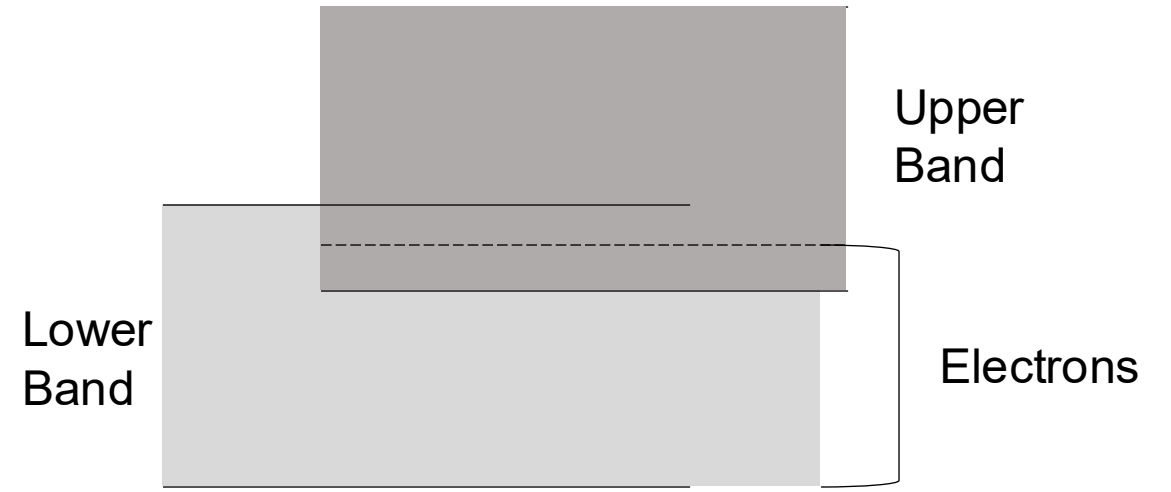
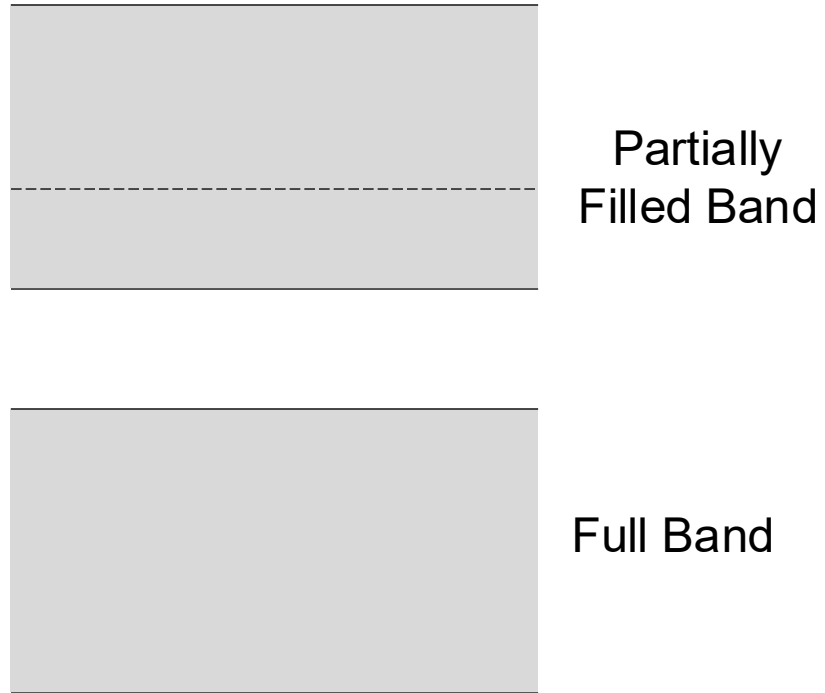


Allowed Energy Band
(Full)

- ✓ Apply electric Field
- ✓ Free holes
- ✓ Current



Metals, Insulators and Semiconductors



Concept of Density of States

$$J = qNv_d$$

$$J_n = qnv_d$$

$$J_p = qp v_d$$

$n \rightarrow$ no of free electron per unit volume

$p \rightarrow$ no of free electron per unit volume

$n \rightarrow$ can be calculated if we know how many states are filled in CB
(total number of available states* probability that a state is filled)

$p \rightarrow$ can be calculated if we know how many states are empty in VB
(total number of available states* probability that a state is empty)
(total number of available states* (1-probability that a state is filled))

Density of States: Number of states per unit energy per unit volume

States: Solution of Schrodinger equation

Steps to calculate Density of States

1. How many solution of Schrödinger equation per unit volume in k –space
2. How many states between k and $k + dk$
3. How many states between E and $E + dE$
4. Number of states per unit volume between E and $E + dE$
5. Number of states per unit energy per unit volume

Steps to calculate Density of States

1. How many solution of Schrödinger equation per unit volume in k –space (3D)

Solution of Schrödinger equation

$$\psi = \sqrt{\frac{2}{L}} \sin kx, \quad k = \frac{n\pi}{L}$$

$$\psi_x = \sqrt{\frac{2}{L_x}} \sin k_x x, \quad k_x = \frac{n_x \pi}{L_x}$$

$$\psi_y = \sqrt{\frac{2}{L_y}} \sin k_y y, \quad k_y = \frac{n_y \pi}{L_y}$$

$$\psi_z = \sqrt{\frac{2}{L_z}} \sin k_z z, \quad k_z = \frac{n_z \pi}{L_z}$$

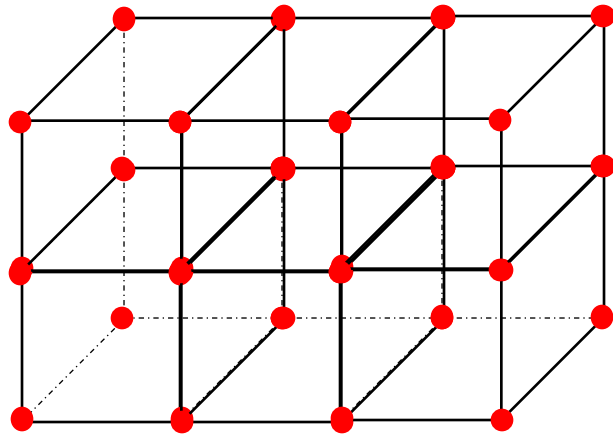

 k can take only quantized value

$$\psi = \psi_x \psi_y \psi_z = \sqrt{\frac{2}{L_x}} \sin k_x x \sqrt{\frac{2}{L_y}} \sin k_y y \sqrt{\frac{2}{L_z}} \sin k_z z$$

Steps to calculate Density of States

1. How many solution of Schrödinger equation per unit volume in k –space (3D)

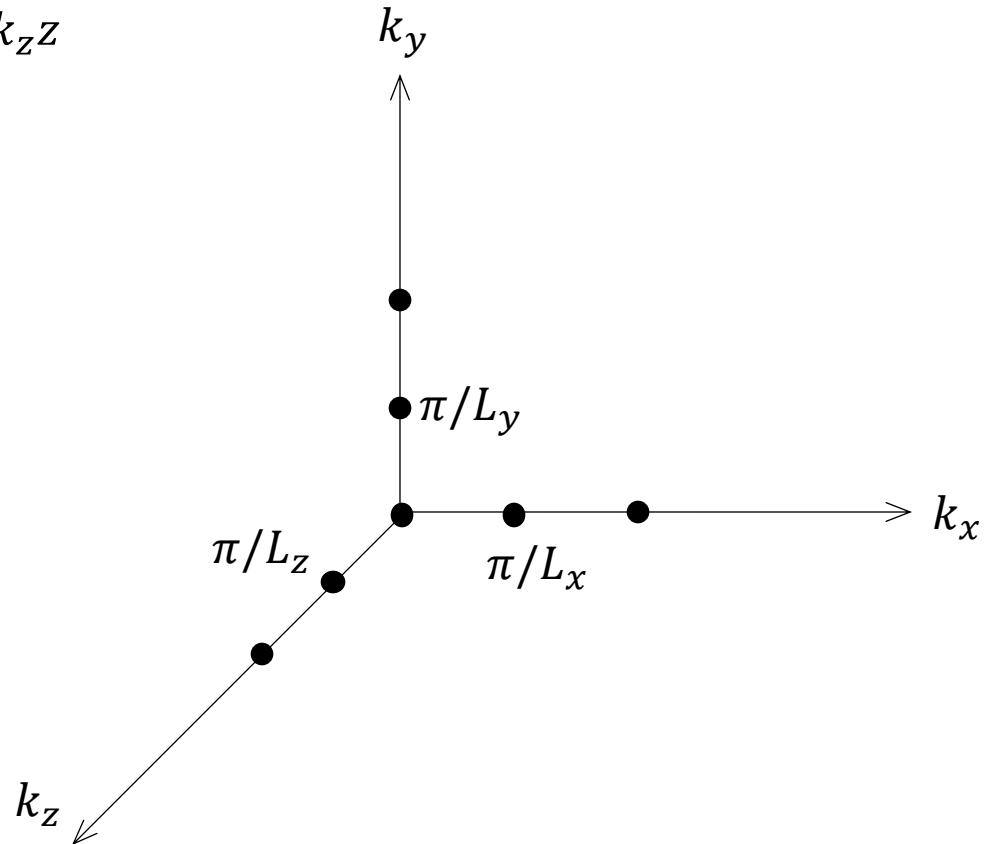
$$\psi = \sqrt{\frac{2}{L_x}} \sqrt{\frac{2}{L_y}} \sqrt{\frac{2}{L_z}} \sin k_x x \sin k_y y \sin k_z z$$



$$\text{Solution per unit cell} = 8 * \frac{1}{8} = 1$$

$$\text{Volume of the unit cell} = \frac{\pi}{L_x} \cdot \frac{\pi}{L_y} \cdot \frac{\pi}{L_z} = \frac{\pi^3}{L_x L_y L_z}$$

$$\text{Solution of Schrödinger equation per unit volume in } k \text{ –space} = \frac{1}{\frac{\pi^3}{L_x L_y L_z}} = \frac{L_x L_y L_z}{\pi^3}$$



Correction Factor

$$k_x = \frac{n_x \pi}{L_x}$$

$$k_y = \frac{n_y \pi}{L_y}$$

$$k_z = \frac{n_z \pi}{L_z}$$

In the above equations n_x can take values ± 1 which will lead to same state: correction factor = $\frac{1}{2}$

Considering all x, y and z direction: correction factor = $\frac{1}{2} * \frac{1}{2} * \frac{1}{2} = \frac{1}{8}$

Electrons can have both up and down spin which can be present at the same energy level

correction factor = 2

$$\text{Final correction factor} = \frac{1}{8} * 2 = \frac{1}{4}$$

Solution of Schrödinger equation per unit volume in k –space = $\frac{L_x L_y L_z}{4\pi^3}$

2. How many states between k and $k + dk$

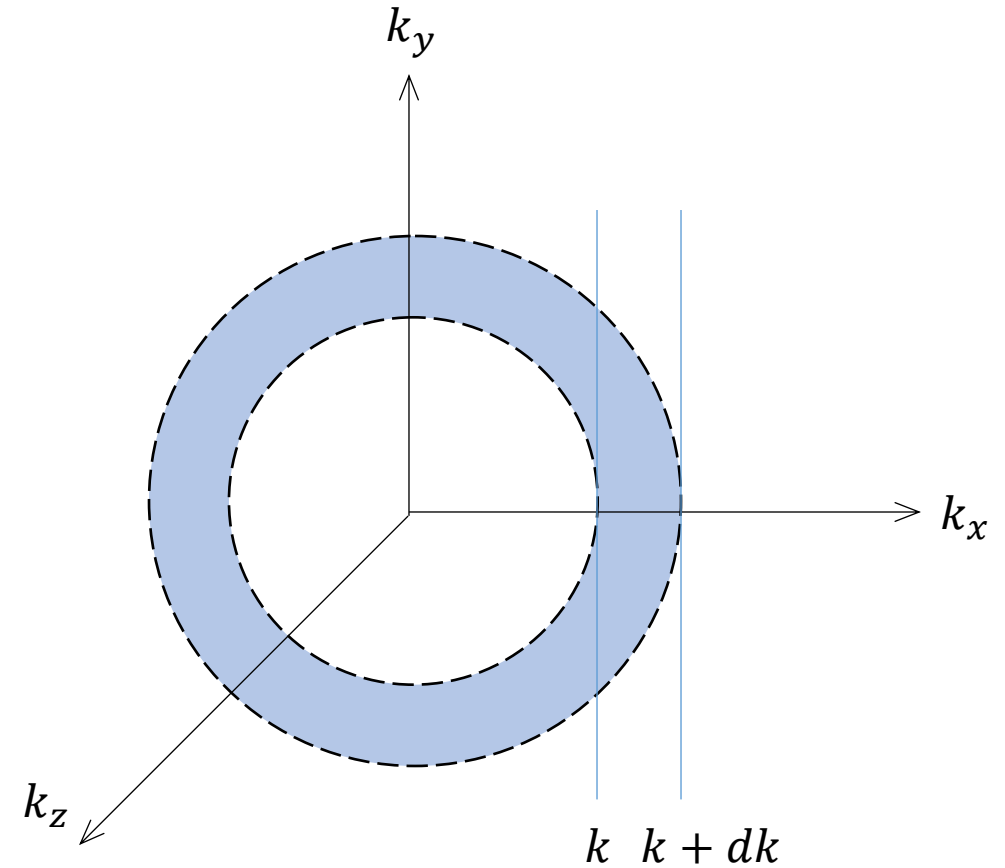
Volume of the space between k and $k + dk$ in 3D

$$4\pi k^2 dk$$

States between k and $k + dk$

$$\frac{L_x L_y L_z}{4\pi^3} * 4\pi k^2 dk$$

$$4\pi k^2 dk * \frac{L_x L_y L_z}{4\pi^3}$$

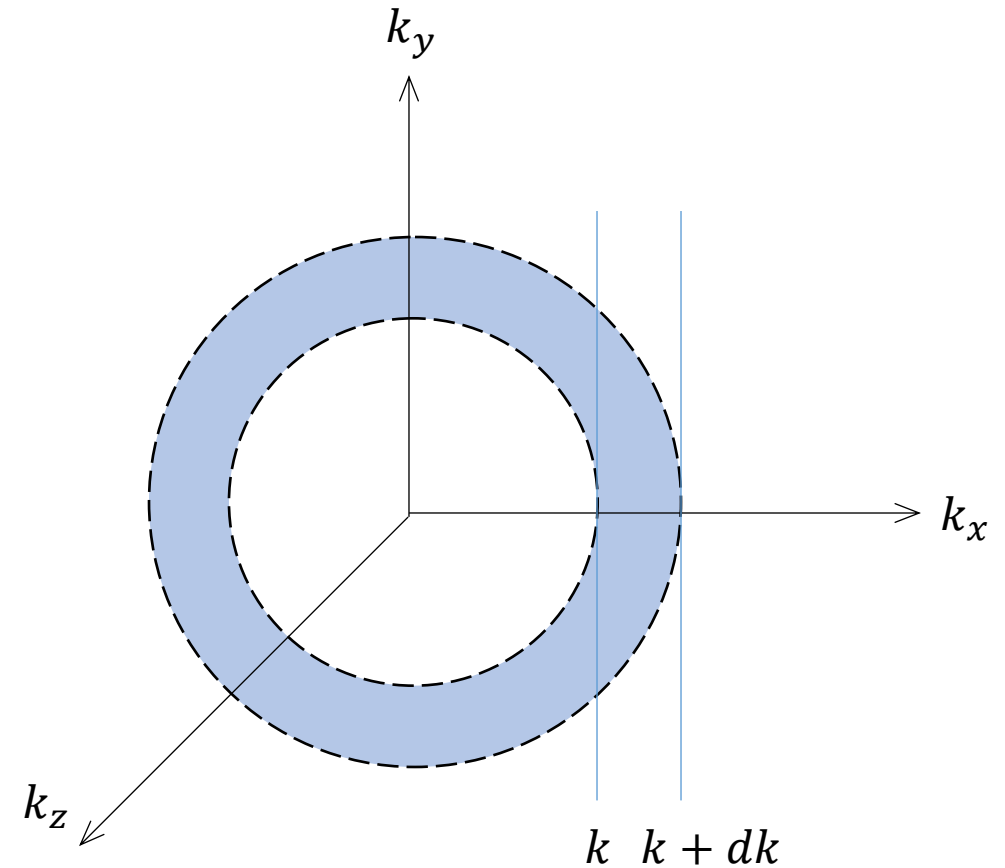


Clarification regarding volume

$$\begin{aligned} V &= \frac{4}{3}\pi(k + dk)^3 - \frac{4}{3}\pi k^3 \\ &= \frac{4}{3}\pi[k^3 + 3k^2dk + 3k(dk)^2 + (dk)^3] - \frac{4}{3}\pi k^3 \\ &= \frac{4}{3}\pi k^3 + 4\pi k^2dk + 4\pi k(dk)^2 + \frac{4}{3}\pi(dk)^3 - \frac{4}{3}\pi k^3 \\ &= 4\pi k^2dk + 4\pi k(dk)^2 + \frac{4}{3}\pi(dk)^3 \\ &= 4\pi k^2dk \left[1 + \frac{4\pi k(dk)^2}{4\pi k^2dk} + \frac{\frac{4}{3}\pi(dk)^3}{4\pi k^2dk} \right] \\ &= 4\pi k^2dk \left[1 + \frac{dk}{k} + \frac{(dk)^2}{3k^2} \right] \end{aligned}$$

If $dk \ll k$

$$V = 4\pi k^2dk$$



$$4\pi k^2 dk * \frac{L_x L_y L_z}{4\pi^3}$$

3. How many states between E and $E + dE$

$$E = \frac{\hbar^2 k^2}{2m}$$

$$k^2 = \frac{2mE}{\hbar^2}$$

$$k = \frac{\sqrt{2mE}}{\hbar}$$

$$dk = \frac{\sqrt{2m}}{\hbar} d(\sqrt{E})$$

$$dk = \frac{\sqrt{2m}}{\hbar} d(\sqrt{E})$$

$$dk = \frac{\sqrt{2m}}{\hbar} \frac{1}{2} E^{-\frac{1}{2}} dE$$

States between E and $E + dE$

$$4\pi \frac{2mE}{\hbar^2} \frac{\sqrt{2m}}{\hbar} \frac{1}{2} E^{-\frac{1}{2}} dE * \frac{L_x L_y L_z}{4\pi^3}$$

$$= \frac{mE}{\hbar^2} \frac{\sqrt{2m}}{\hbar} E^{-\frac{1}{2}} dE * \frac{L_x L_y L_z}{\pi^2}$$

$$= L_x L_y L_z \frac{\sqrt{2} m^{3/2} E^{1/2}}{\pi^2 \hbar^3} dE$$

$$= L_x L_y L_z \frac{(2m^3)^{1/2} E^{1/2}}{\pi^2 \hbar^3} dE$$

4. Number of states per unit volume between E and $E + dE$

$$= \frac{(2m^3)^{1/2} E^{1/2}}{\pi^2 \hbar^3} dE$$

5. Number of states per unit energy per unit volume

$$= \frac{(2m^3)^{1/2} E^{1/2}}{\pi^2 \hbar^3}$$

$$DOS = \frac{(2m^3)^{1/2} E^{1/2}}{\pi^2 \hbar^3} \propto E^{1/2}$$

Energy space volume

$$V = L_x * L_y * L_z$$

k – space volume

$$\frac{\pi}{L_x} \cdot \frac{\pi}{L_y} \cdot \frac{\pi}{L_z} = \frac{\pi^3}{L_x L_y L_z}$$

Conduction Band

$$E - E_c = C_1 k^2$$

$$E - E_c = \frac{\hbar^2 k^2}{2m_e^*}$$

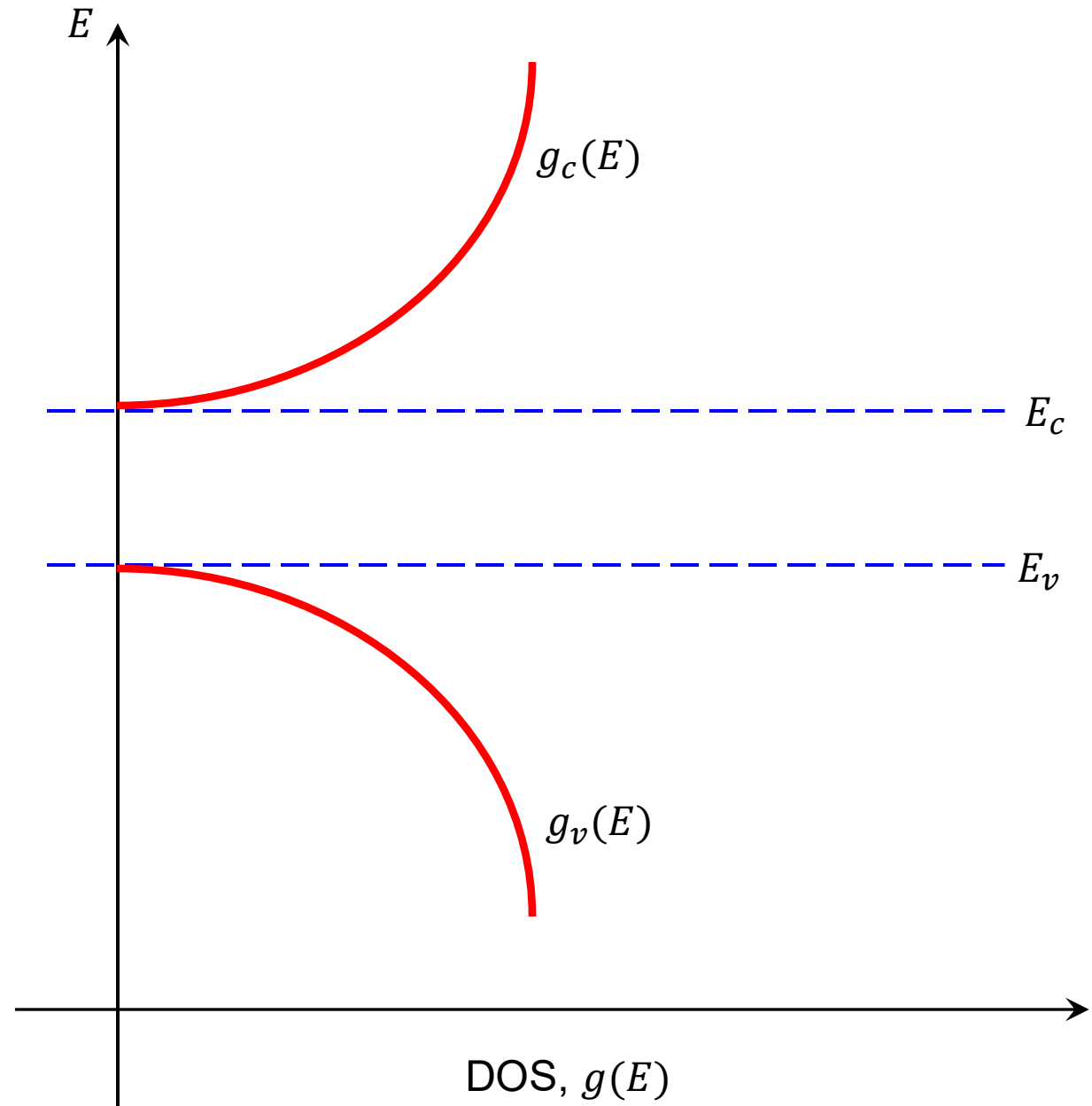
$$DOS = \frac{(2m^3)^{1/2}}{\pi^2 \hbar^3} \sqrt{E - E_c}$$

Valence Band

$$E_v - E = C_2 k^2$$

$$E_v - E = \frac{\hbar^2 k^2}{2m_h^*}$$

$$DOS = \frac{(2m^3)^{1/2}}{\pi^2 \hbar^3} \sqrt{E_v - E}$$



Density of States (2D)

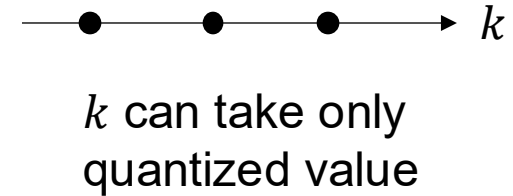
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Solution of Schrödinger equation

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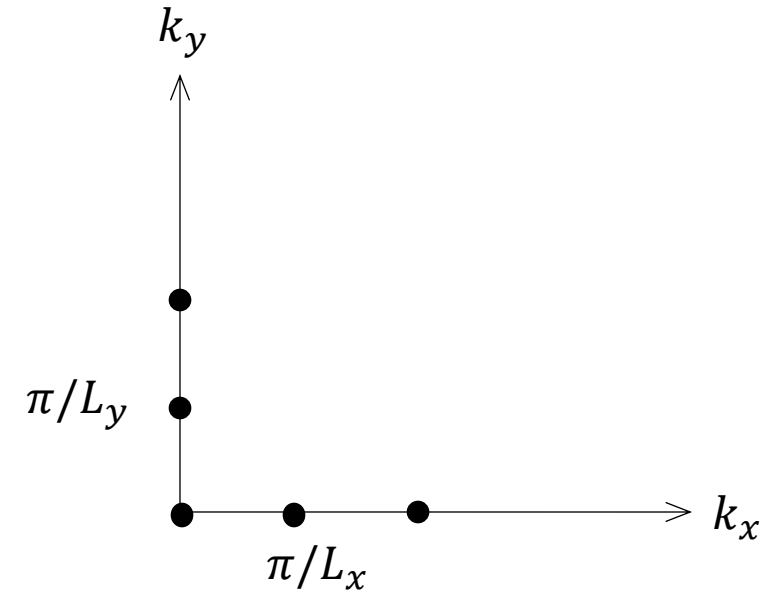
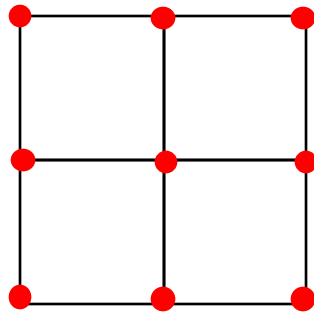


$$\psi = \psi_x \psi_y \psi_z = \sqrt{\frac{2}{L_x}} \sin k_x x \sqrt{\frac{2}{L_y}} \sin k_y y$$

Steps to calculate Density of States

1. How many solution of Schrödinger equation per unit area in k –space (3D)

$$\psi = \sqrt{\frac{2}{L_x}} \sqrt{\frac{2}{L_y}} \sqrt{\frac{2}{L_z}} \sin k_x x \sin k_y y \sin k_z z$$



$$\text{Solution per unit area} = 4 * \frac{1}{4} = 1$$

$$\text{Area of one cell} = \frac{\pi}{L_x} \cdot \frac{\pi}{L_y} = \frac{\pi^2}{L_x L_y}$$

$$\text{Solution of Schrödinger equation per unit area in } k \text{ –space} = \frac{1}{\frac{\pi^2}{L_x L_y}} = \frac{L_x L_y}{\pi^2}$$

Correction Factor

$$k_x = \frac{n_x \pi}{L_x} \quad k_y = \frac{n_y \pi}{L_y}$$

In the above equations n_x can take values ± 1 which will lead to same state: correction factor = $\frac{1}{2}$

Considering all x, y and z direction: correction factor = $\frac{1}{2} * \frac{1}{2} = \frac{1}{4}$

Electrons can have both up and down spin which can be present at the same energy level

correction factor = 2

$$\text{Final correction factor} = \frac{1}{4} * 2 = \frac{1}{2}$$

Solution of Schrödinger equation per unit area in k –space = $\frac{L_x L_y}{2\pi^2}$

2. How many states between k and $k + dk$

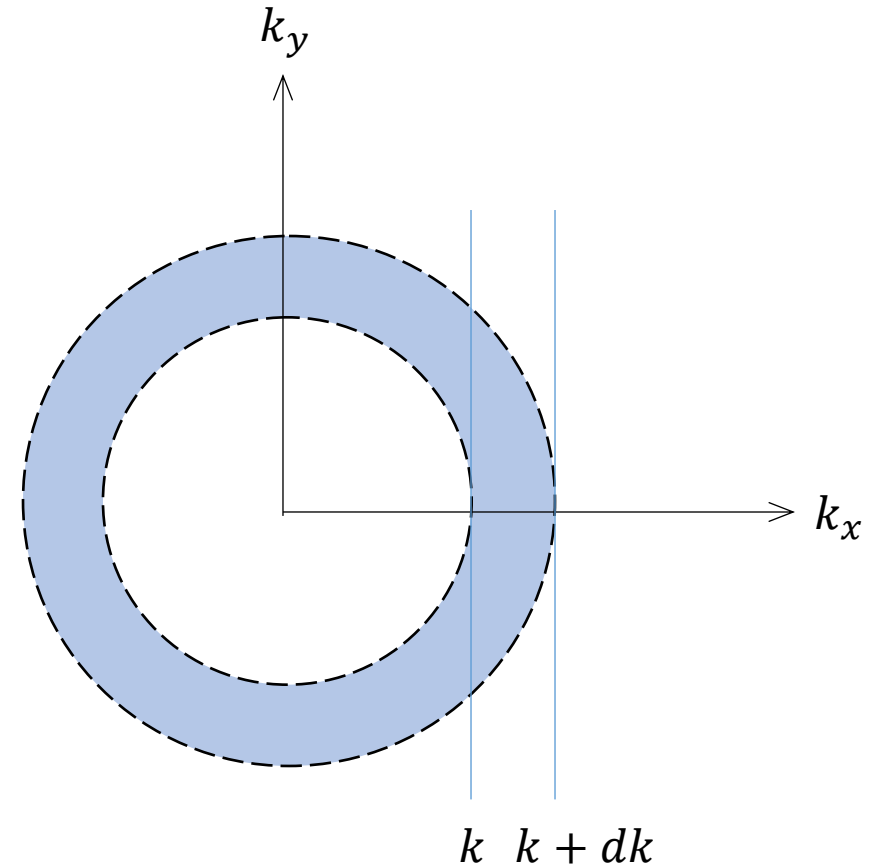
area of the space between k and $k + dk$ in 2D

$$2\pi k dk$$

States between k and $k + dk$

$$\frac{L_x L_y}{2\pi^2} * 2\pi k dk$$

$$2\pi k dk * \frac{L_x L_y}{2\pi^2}$$



$$2\pi k dk * \frac{L_x L_y}{2\pi^2}$$

3. How many states between E and $E + dE$

$$E = \frac{\hbar^2 k^2}{2m}$$

$$k^2 = \frac{2mE}{\hbar^2}$$

$$k = \frac{\sqrt{2mE}}{\hbar}$$

$$dk = \frac{\sqrt{2m}}{\hbar} d(\sqrt{E})$$

$$dk = \frac{\sqrt{2m}}{\hbar} d(\sqrt{E})$$

$$dk = \frac{\sqrt{2m}}{\hbar} \frac{1}{2} E^{-\frac{1}{2}} dE$$

States between E and $E + dE$

$$2\pi \frac{\sqrt{2mE}}{\hbar} \frac{\sqrt{2m}}{\hbar} \frac{1}{2} E^{-\frac{1}{2}} dE * \frac{L_x L_y}{2\pi^2}$$

$$= L_x L_y \frac{m}{\pi \hbar^2} dE$$

4. Number of states per unit area between E and $E + dE$

$$= \frac{m}{\pi \hbar^2} dE$$

5. Number of states per unit energy per unit area

$$= \frac{m}{\pi \hbar^2}$$

Energy space volume

$$V = L_x * L_y$$

k – space volume

$$\frac{\pi}{L_x} \cdot \frac{\pi}{L_y} = \frac{\pi^2}{L_x L_y}$$

Density of States (1D)

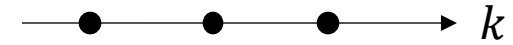
1. How many solution of Schrödinger equation per unit volume in k –space (1D)

Solution of Schrödinger equation

$$\psi_x = \sqrt{\frac{2}{L_x}} \sin k_x x, \quad k_x = \frac{n_x \pi}{L_x}$$

Solution per line length = 1

$$\text{line length} = \frac{\pi}{L_x}$$



k can take only
quantized value

Solution of Schrödinger equation per unit length in k –space = $\frac{L_x}{\pi}$

Correction Factor

$$k_x = \frac{n_x \pi}{L_x}$$

In the above equations n_x can take values ± 1 which will lead to same state: correction factor = $\frac{1}{2}$

Electrons can have both up and down spin which can be present at the same energy level

correction factor = 2

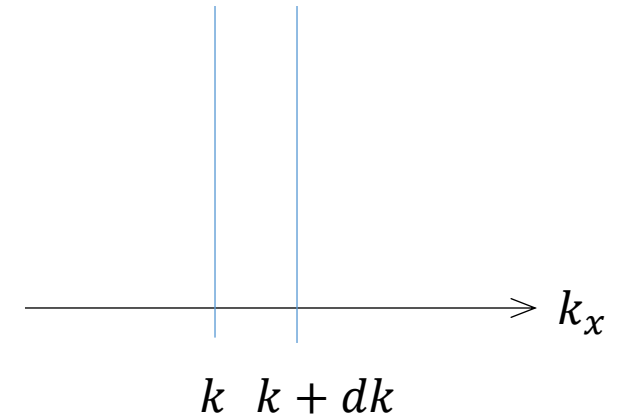
$$\text{Final correction factor} = \frac{1}{2} * 2 = 1$$

Solution of Schrödinger equation per unit length in k –space = $\frac{L_x}{\pi}$

2. How many states between k and $k + dk$

space between k and $k + dk$ in 1D: dk

States between k and $k + dk$: $\frac{L_x}{\pi} dk$



3. How many states between E and $E + dE$

$$k = \frac{\sqrt{2mE}}{\hbar}$$

States between E and $E + dE$

$$dk = \frac{\sqrt{2m}}{\hbar} d(\sqrt{E})$$

$$\frac{L_x}{\pi} dk$$

$$dk = \frac{\sqrt{2m}}{\hbar} \frac{1}{2} E^{-\frac{1}{2}} dE$$

$$\frac{L_x}{\pi} \frac{\sqrt{2m}}{\hbar} \frac{1}{2} E^{-\frac{1}{2}} dE$$

4. Number of states per unit length between E and $E + dE$

$$\frac{1}{\pi} \frac{\sqrt{2m}}{\hbar} \frac{1}{2} E^{-\frac{1}{2}} dE$$

5. Number of states per unit energy per unit length

$$\frac{1}{\pi} \frac{\sqrt{2m}}{\hbar} \frac{1}{2} E^{-\frac{1}{2}}$$

Fermi-Dirac probability function

The **Fermi-Dirac probability function** describes the likelihood that an electronic energy state is occupied by a **fermion** (typically an **electron**) at a given temperature. It is essential in **semiconductor physics**, **solid-state physics**, and **quantum statistics**.

Fermi-Dirac Distribution Function

The function is given by:

$$f(E) = \frac{1}{1 + \exp(\frac{E - E_F}{KT})}$$

Where:

$f(E)$: Probability that an energy state at energy E is occupied.

E : Energy level of the state.

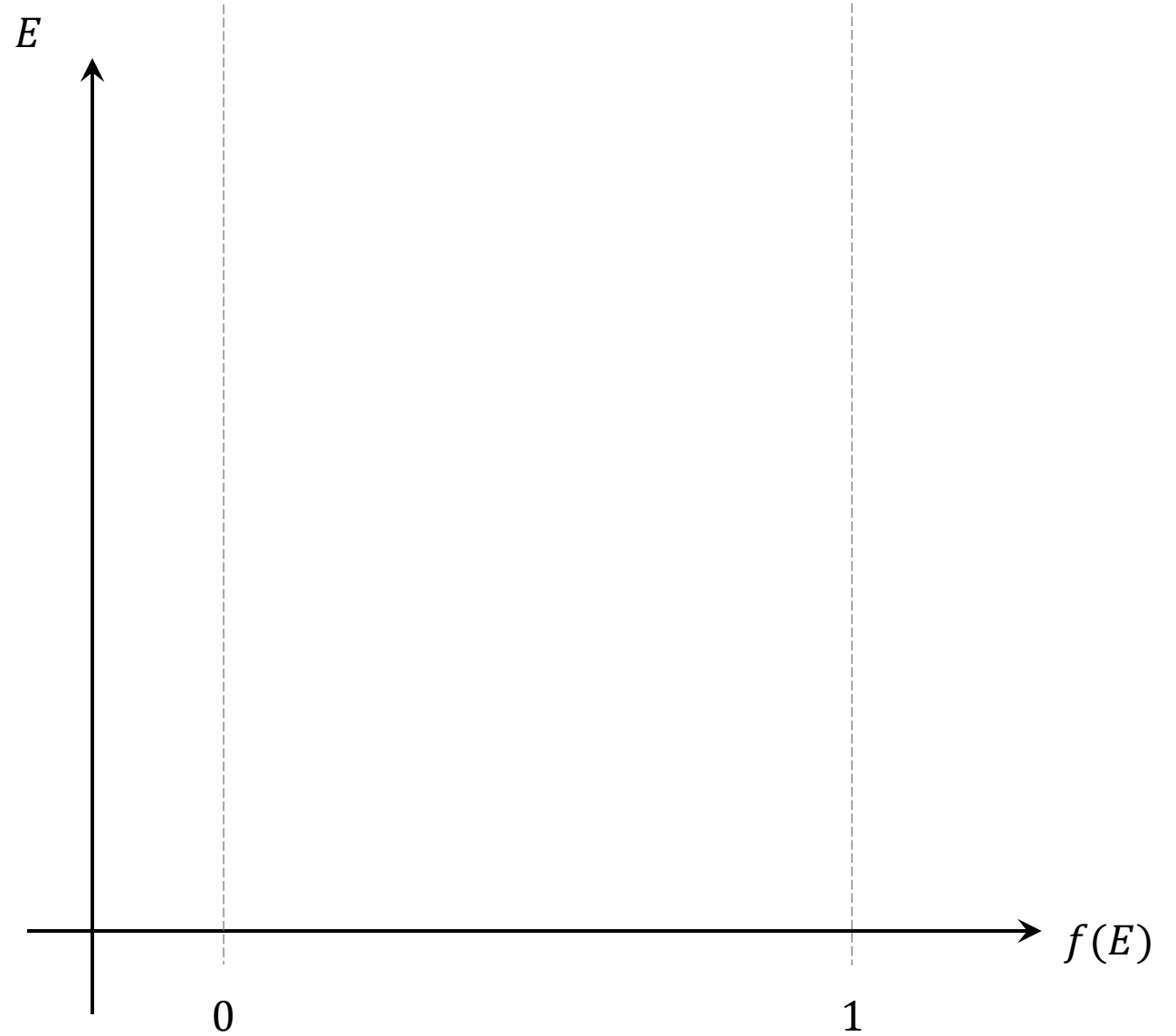
E_F : **Fermi energy**, the energy at which the probability of occupancy is 50% when $T = 0K$.

K : Boltzmann constant ($\approx 8.617 \times 10^{-5} \text{ eV/K}$)

T : Absolute temperature in K

Fermi-Dirac probability function

$$f(E) = \frac{1}{1 + \exp(\frac{E - E_F}{KT})}$$



Fermi-Dirac probability function

$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{KT}\right)}$$

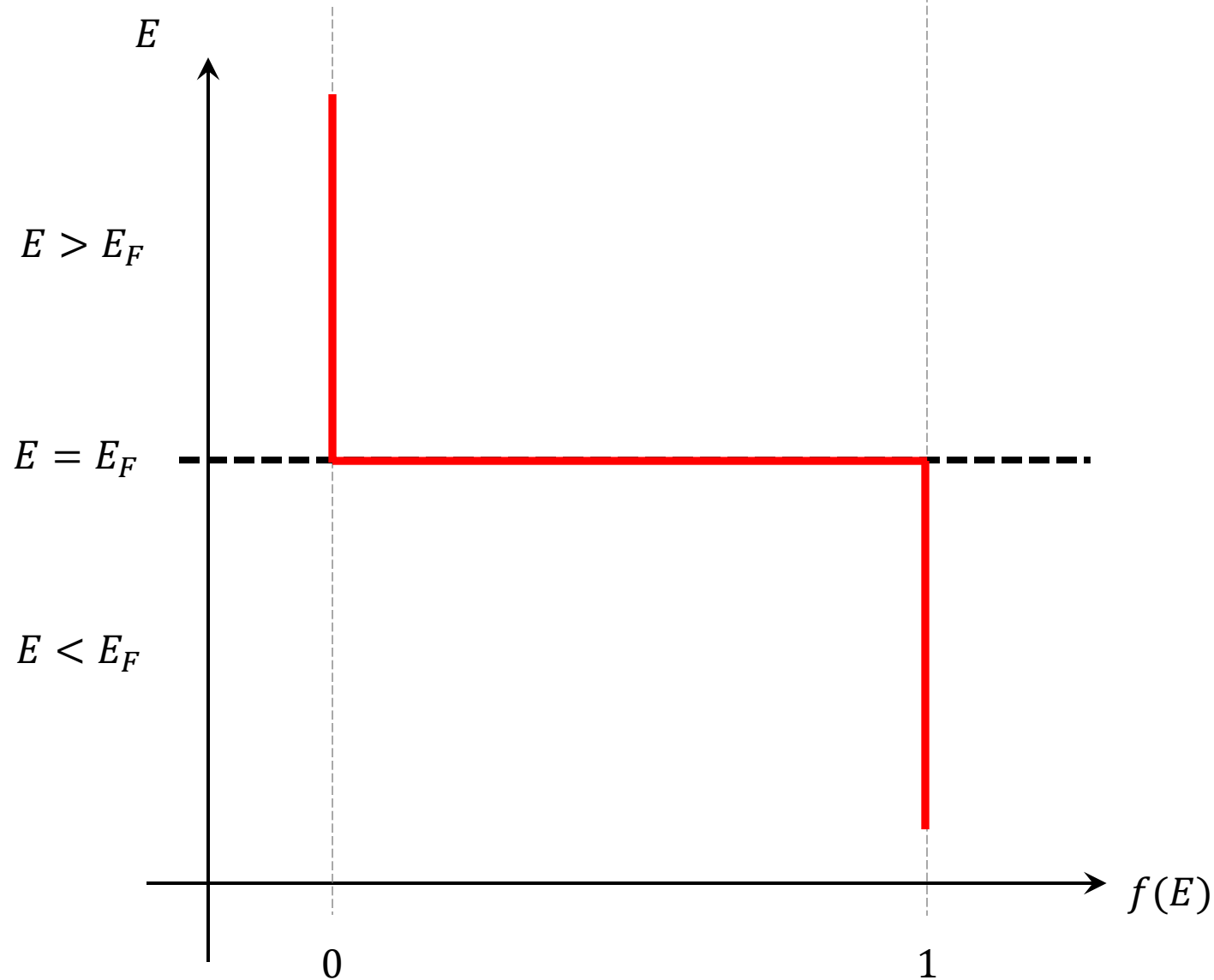
At $T = 0K$

If $E > E_F$

$$f(E) = \frac{1}{1 + \exp(\infty)} = 0$$

If $E < E_F$

$$f(E) = \frac{1}{1 + \exp(-\infty)} = 1$$



Fermi-Dirac probability function

$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{KT}\right)}$$

At $T \neq 0K$

If $E = E_F$

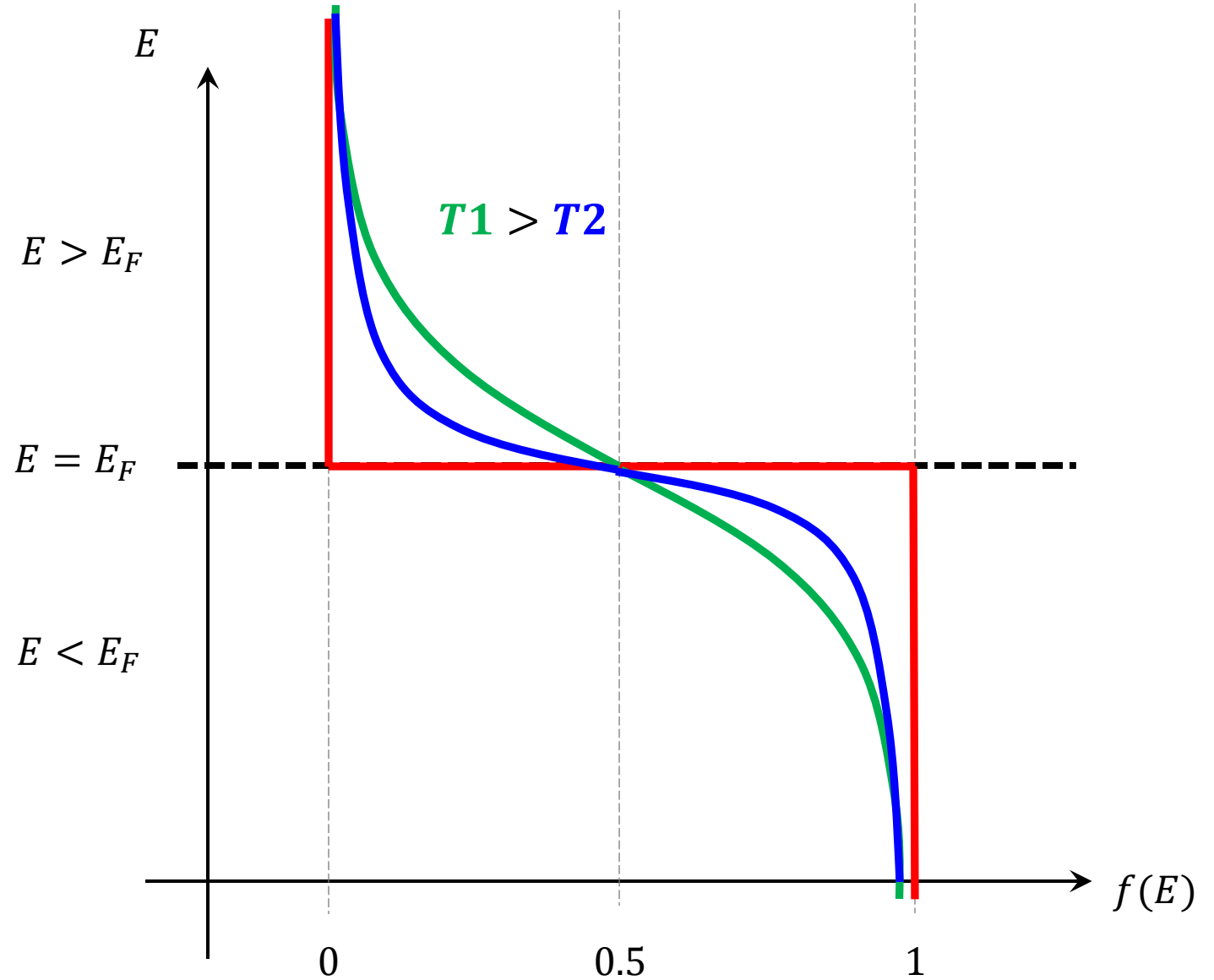
$$f(E) = \frac{1}{1 + \exp(0)} = \frac{1}{2}$$

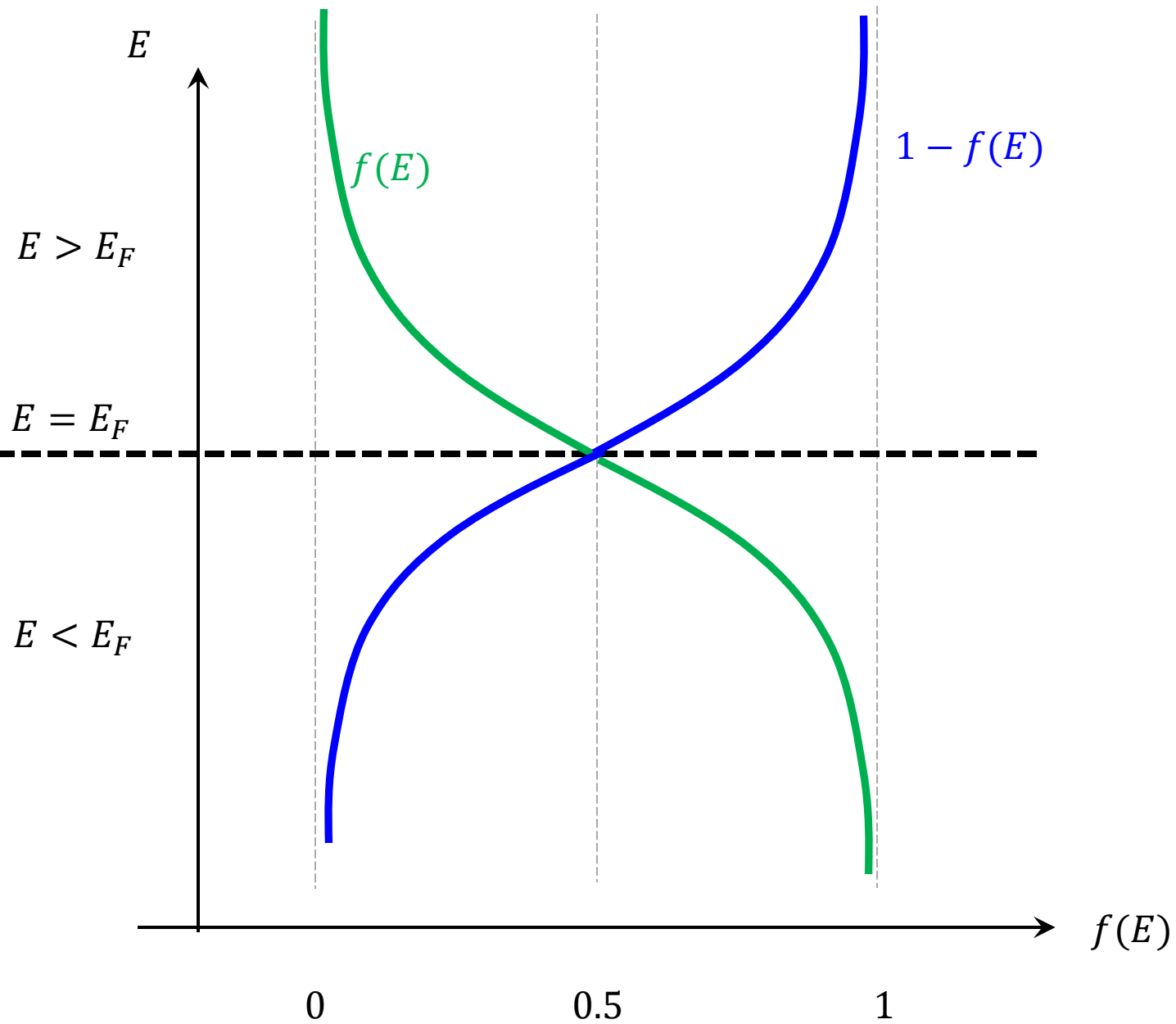
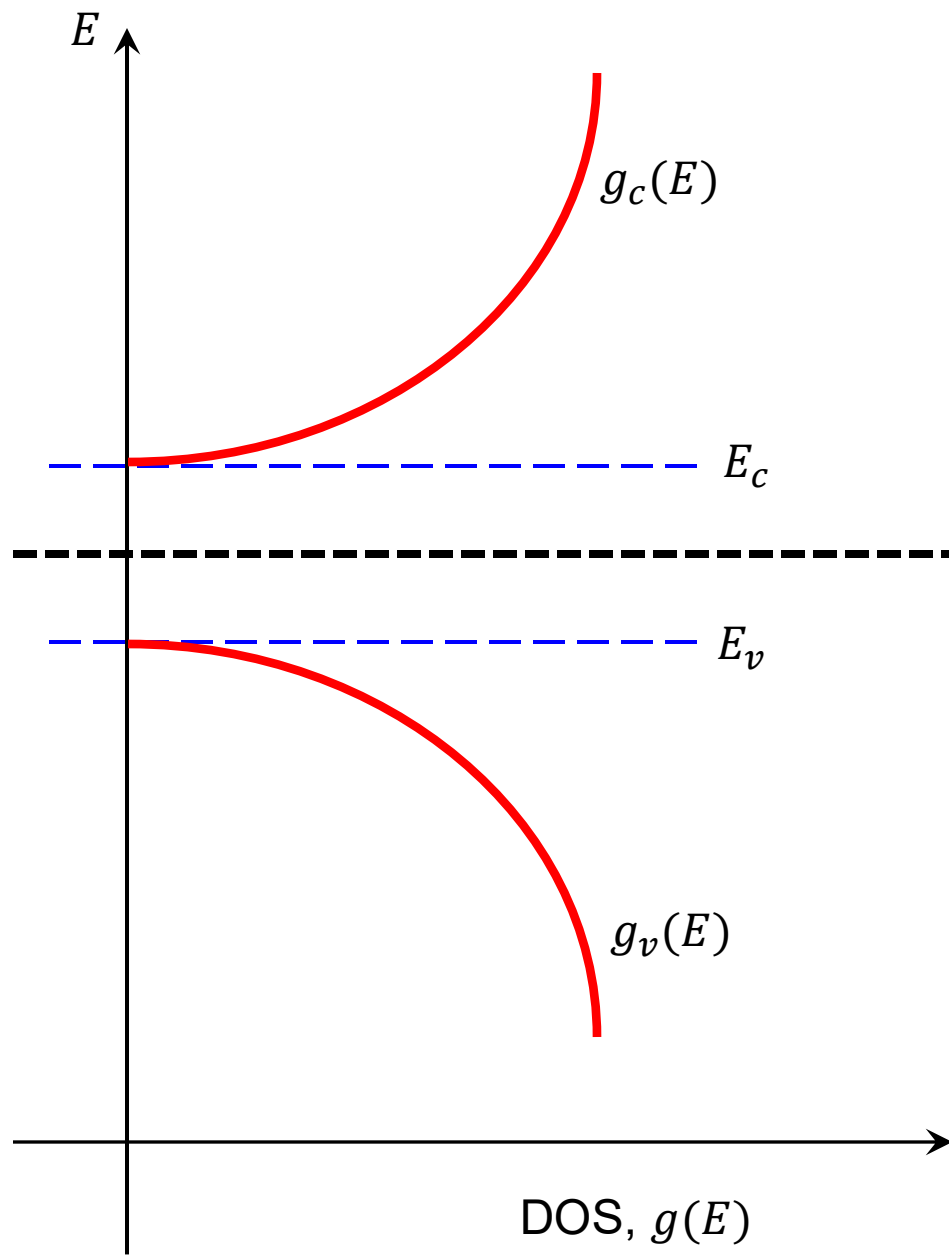
If $E \rightarrow \infty$

$$f(E) = \frac{1}{1 + \exp(\infty)} = 0$$

If $E \rightarrow -\infty$

$$f(E) = \frac{1}{1 + \exp(-\infty)} = 1$$

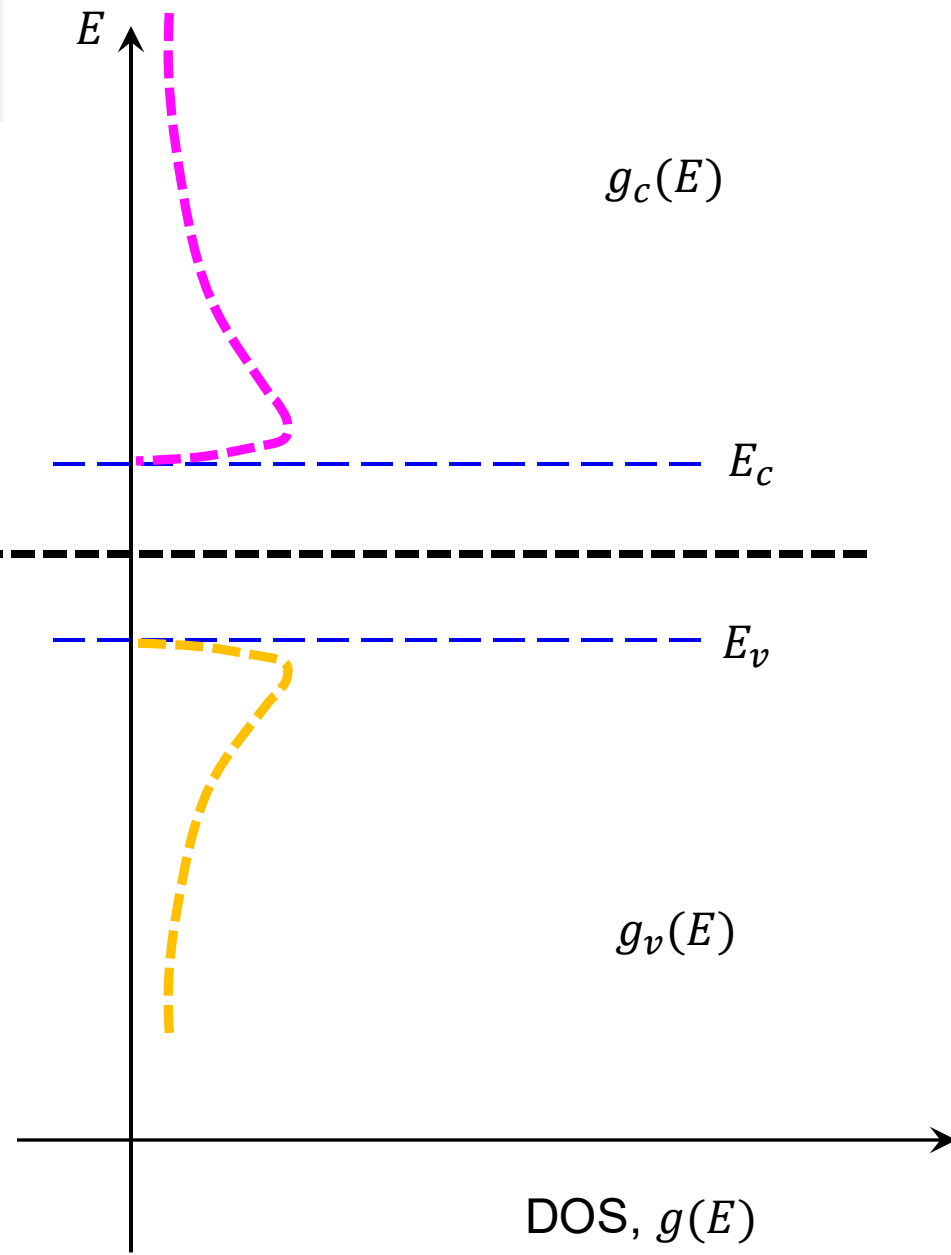
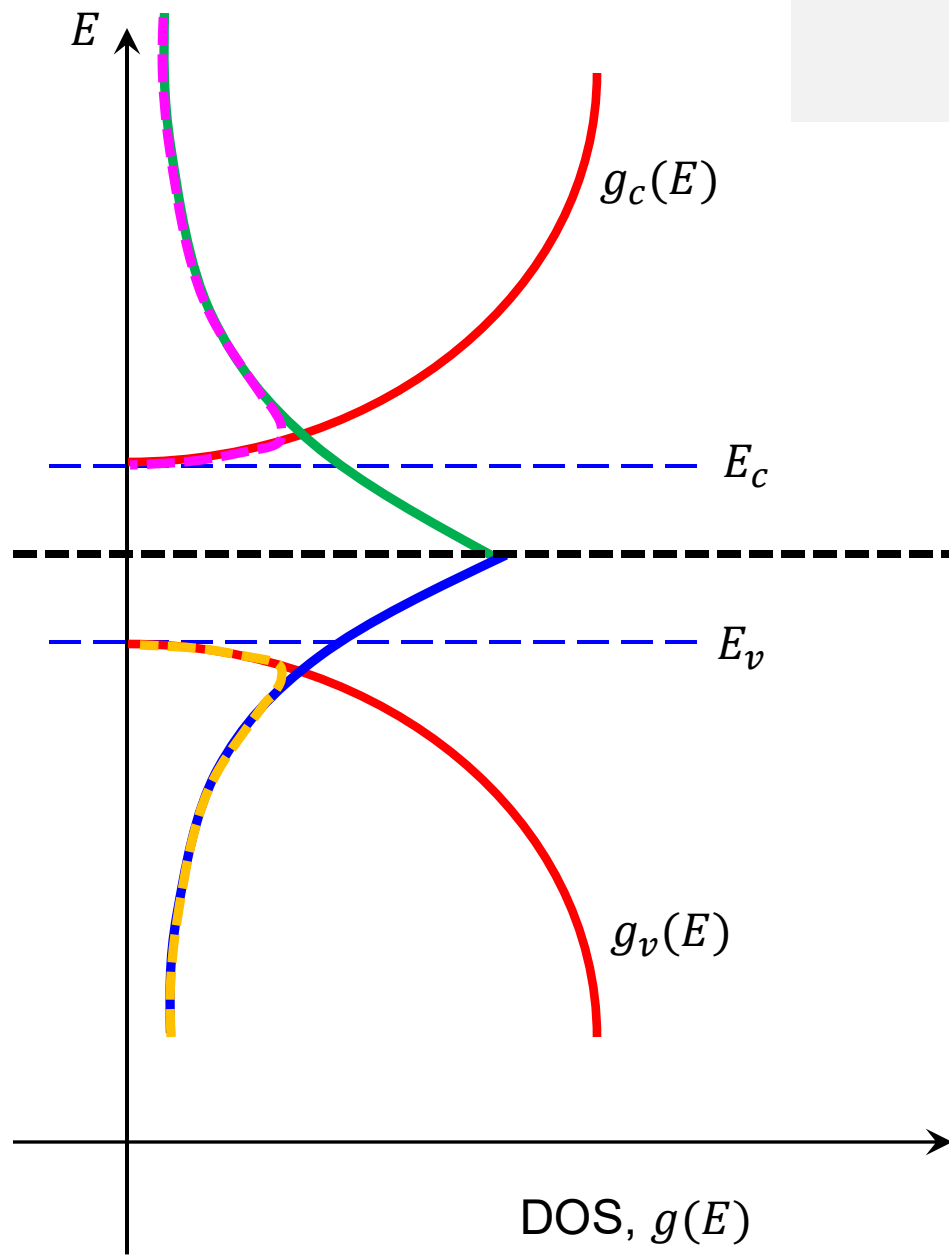




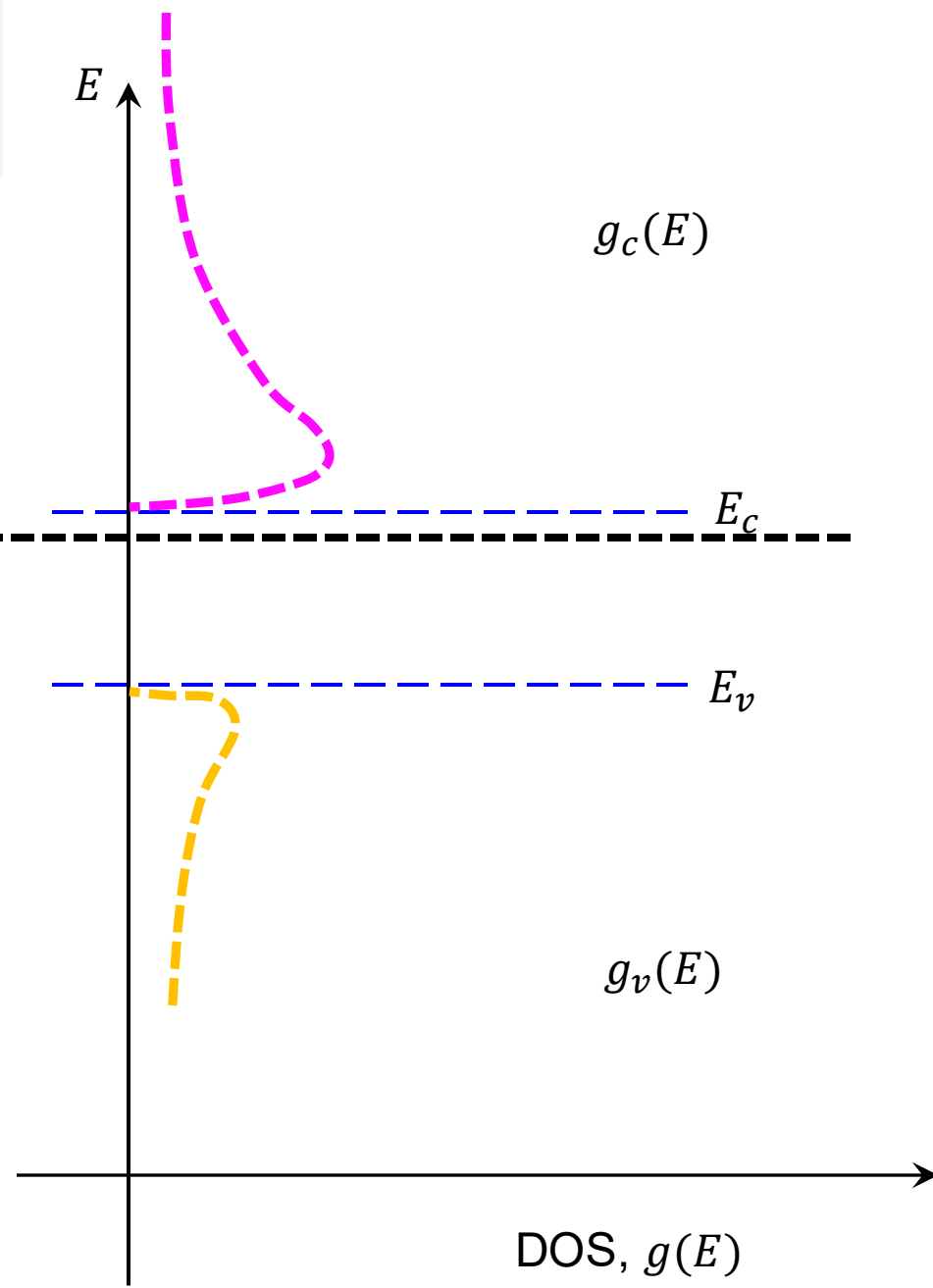
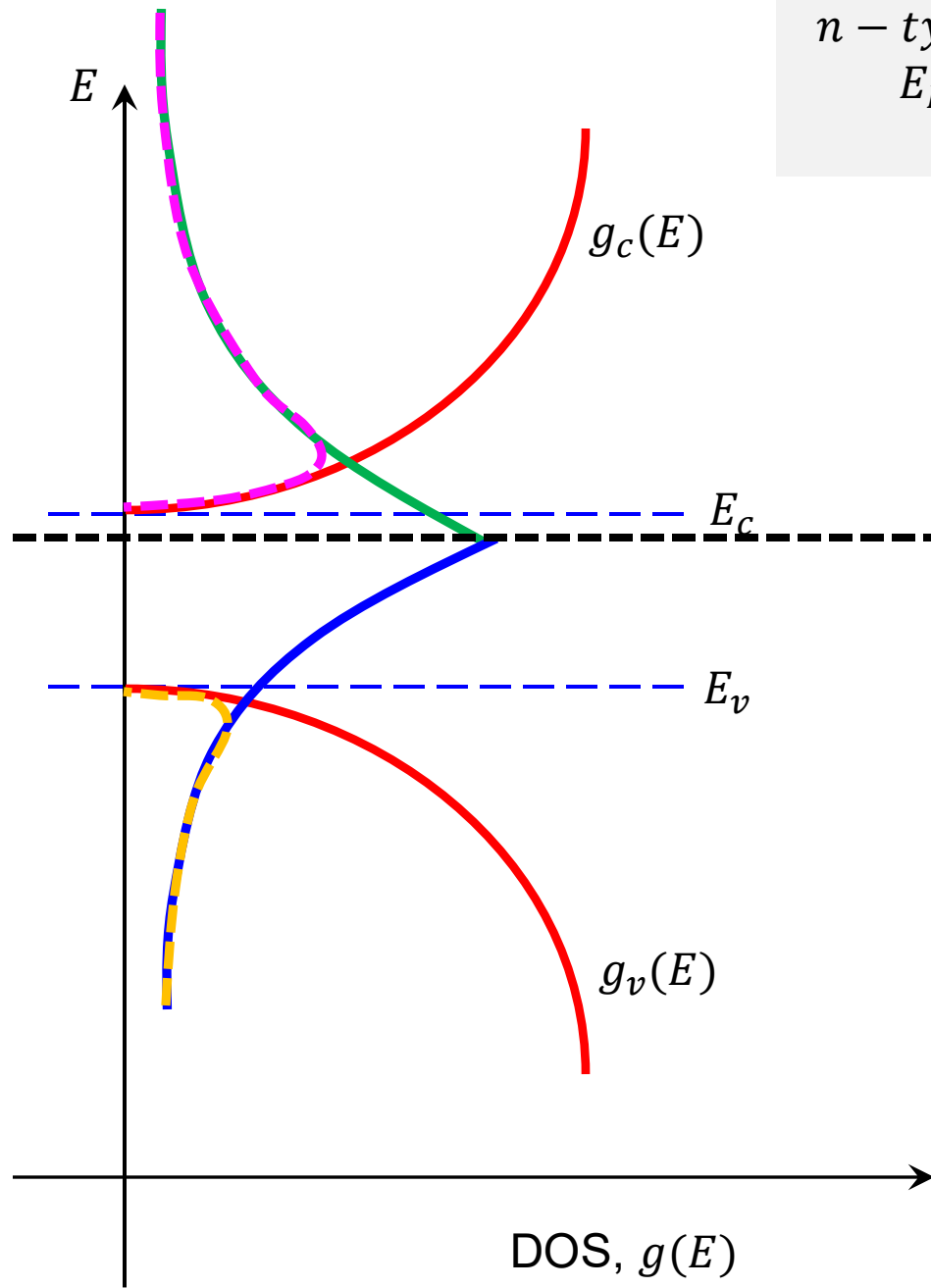
Intrinsic semiconductor

$E_F \sim \text{centre}$

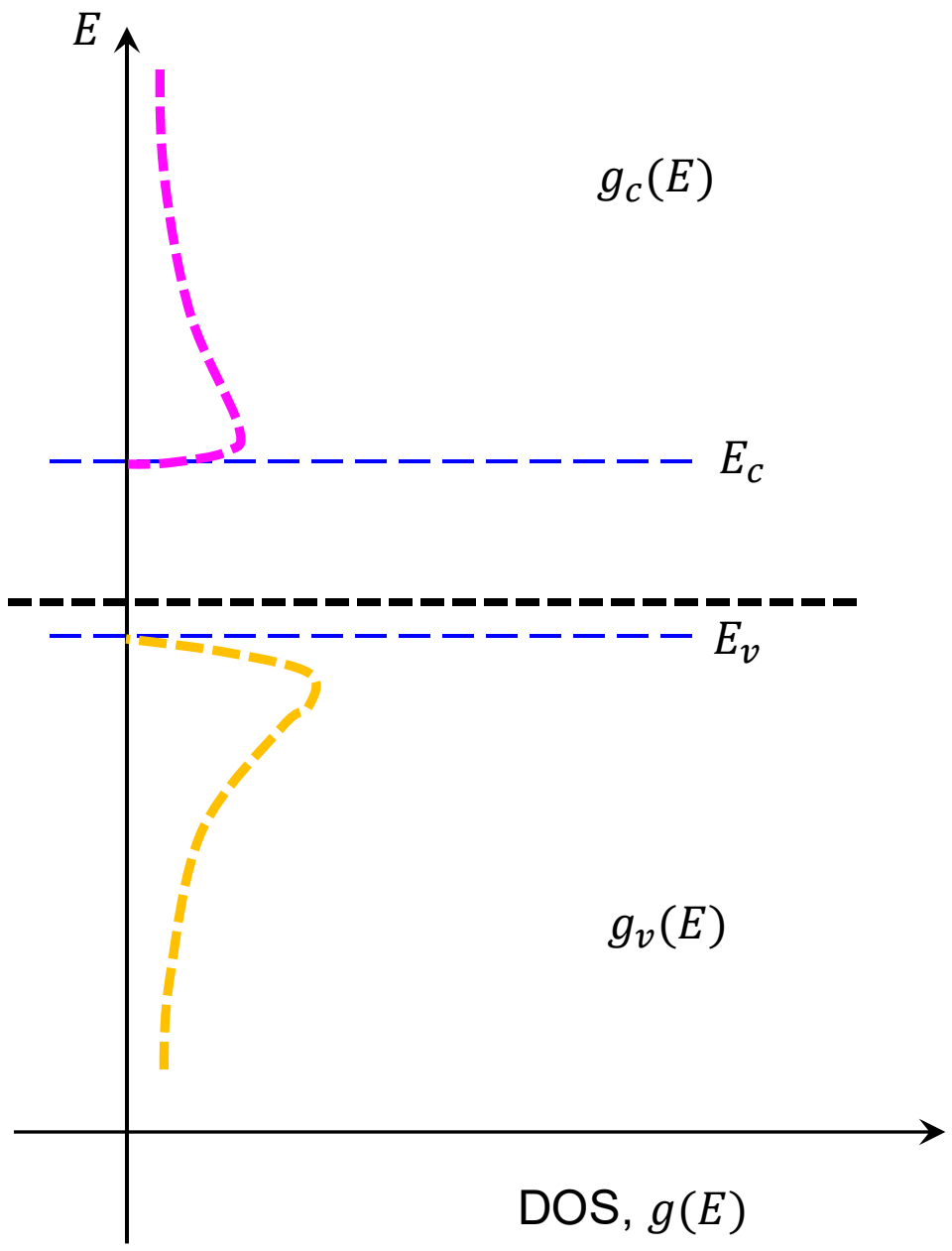
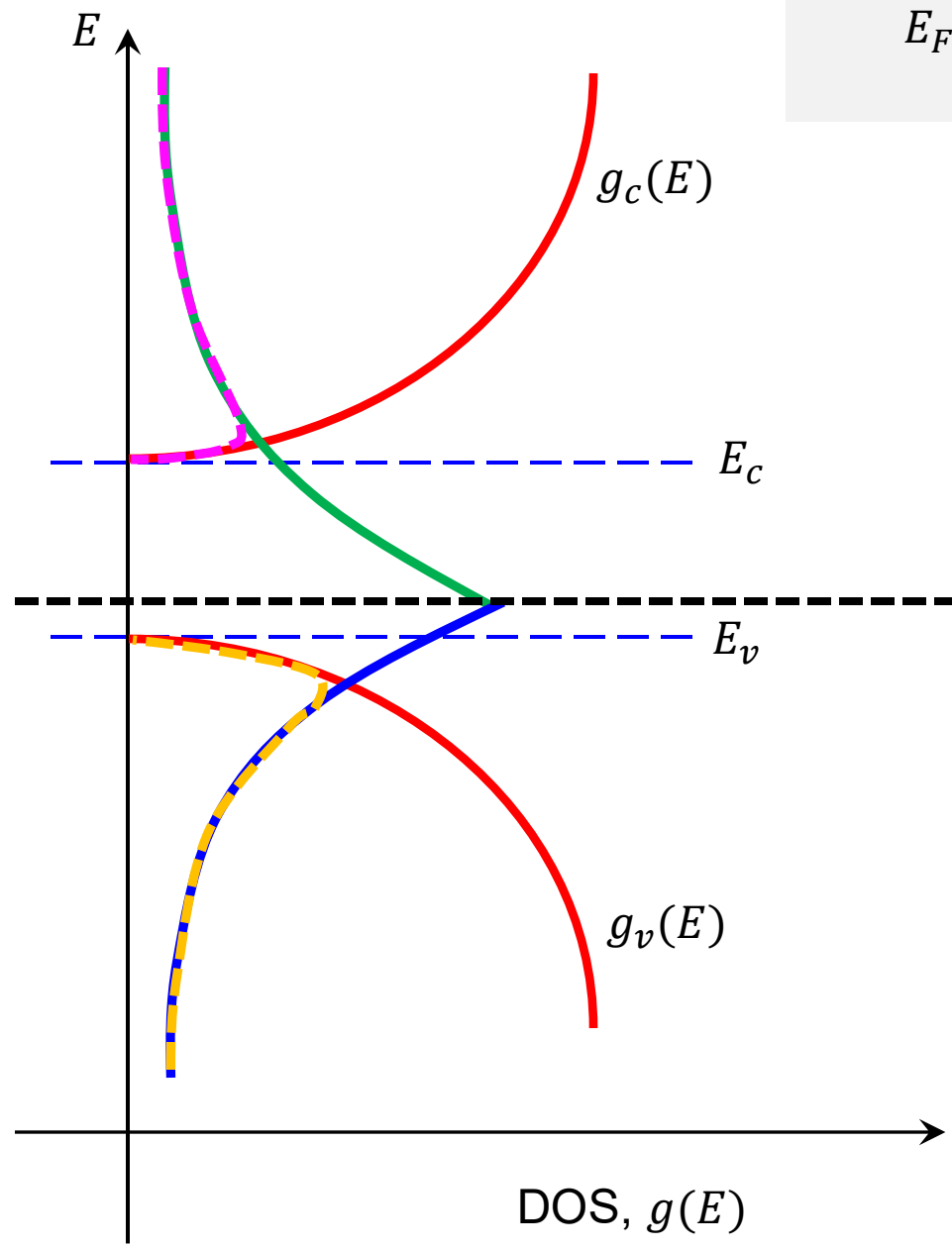
$n = p$



n – type semiconductor
 $E_F \sim$ close to E_c
 $n \gg p$



p – type semiconductor
 $E_F \sim$ close to E_v
 $p \gg n$

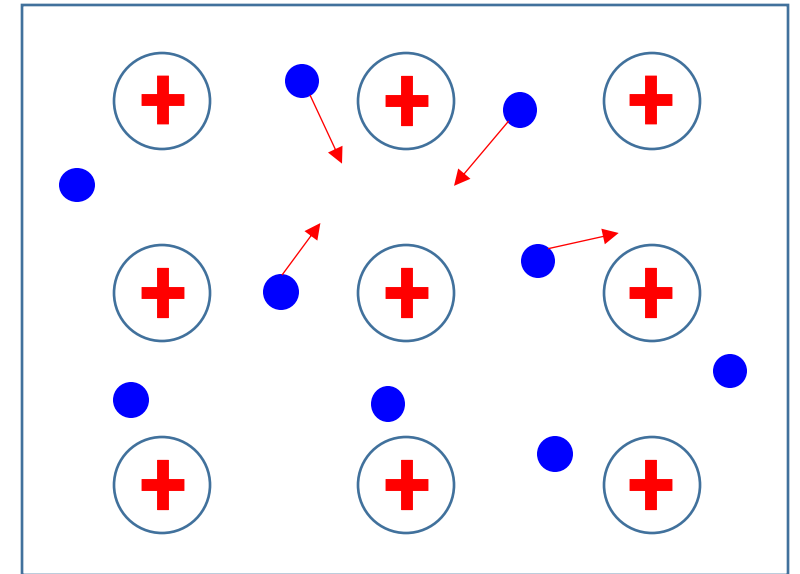


Drude Model/ Free electron gas model

- In metals there are **large number of free electrons** moving freely in all possible directions. Free electrons in the metals are assumed to behave like **gas molecules obeying the kinetic theory of gases**. The free electrons are therefore referred to as free electron gas and the model is also named as Free electron gas model.
- Electrons undergo collisions with each other or fixed ion cores. All these collisions are elastic in nature i.e. there is no loss of energy.
- The interaction among free electrons, electron and ionic core is neglected. As a result, free electrons experience a constant zero potential within the metal.
- The free electrons will obey BOLTZMAN MAXWELL statistics
- In the absence of electric field the kinetic energy of the electrons will be

$$\frac{1}{2}mv^2 = \frac{3}{2}kT$$

- When electric field is applied to the metal the free electrons accelerated in the opposite direction to the applied field.



Drude Model

The electric current density J is defined as the net amount of charge flowing across a unit area per unit time

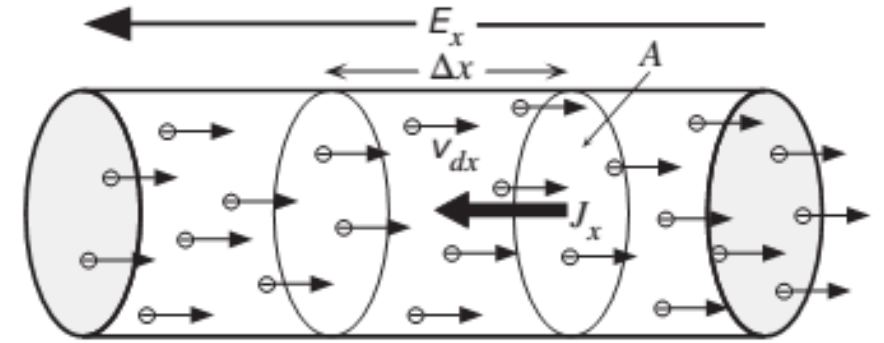
$$J = \frac{I}{A} = \frac{\Delta q}{A \cdot \Delta t}$$

Δq is the net quantity of charge flowing through an area A in time Δt .

If there are N electrons, the average drift velocity v_{dx} can be written as,

$$v_{dx} = \frac{1}{N} \sum_{i=1}^N v_{xi}$$

where v_{xi} is the x direction velocity of the i^{th} electron



- Figure shows the net flow of electrons in a conductor section of cross-sectional area A in the presence of an applied field E_x ($-ve$ x - direction)
- As a result of the application of the electric field E_x , all conduction electrons acquire a net velocity in the x direction

Drude Model

$$J = \frac{I}{A} = \frac{\Delta q}{A \cdot \Delta t}$$

Suppose that,

n is the number of electrons per unit volume in the conductor ($n = N/V$)

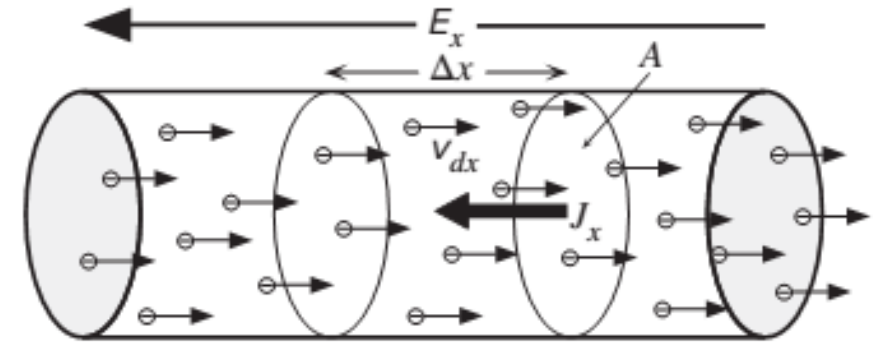
and in time Δt , electrons move a distance Δx
($\Delta x = v_{dx} \Delta t$)

so the total charge Δq crossing the area A is $enA \Delta x$.

The current density in the x direction is

$$J_x = \frac{\Delta q}{A \cdot \Delta t} = \frac{enA \Delta x}{A \cdot \Delta t} = en \frac{\Delta x}{\Delta t} = env_{dx}$$

$$J_x = env_{dx}$$



- Figure shows the net flow of electrons in a conductor section of cross-sectional area A in the presence of an applied field E_x ($-ve$ x - direction)
- As a result of the application of the electric field E_x , all conduction electrons acquire a net velocity in the x direction

Drude Model

Mean Free time or relaxation time:

Let us consider **electrons** (or charge carriers) in a conductor or semiconductor. These electrons move around randomly, but when we apply an **electric field**, they start to move **on average** in a particular direction — this is called **drift velocity**, denoted as $v(t)$

But electrons **don't keep accelerating forever** because they **collide** with atoms or impurities. After each collision, they lose some of their ordered motion and need to build it back up — **this is where the concept of relaxation comes in.**

$$v(t) = v_0 e^{-\frac{t}{\tau}}$$

$$\frac{dv}{dt} = -\frac{v}{\tau}$$

$$a = -\frac{v_{dx}}{\tau}$$

τ is the relaxation time

Mean Free path

$$\lambda = v_{th}\tau$$

v_{th} is called thermal velocity

Drude Model

$$a = -\frac{v_{dx}}{\tau}$$

If electric field E_x is applied on electrons with mass m_e , the force,

$$F = m_e a_x = -eE_x$$

$$a_x = -\frac{eE_x}{m_e} = -\frac{v_{dx}}{\tau}$$

$$v_{dx} = \frac{eE_x\tau}{m_e}$$

$$v_{dx} = \frac{e\tau}{m_e} E_x = \mu E_x$$

$$J_x = en v_{dx} = en \frac{eE_x\tau}{m_e}$$

$$J_x = en v_{dx} = en \mu E_x$$

$$J_x = \frac{e^2 n \tau}{m_e} E_x$$

$$J_x = en \mu E_x$$

$$J_x = \sigma E_x$$

$$\sigma = en \mu$$

$$\sigma = \frac{e^2 n \tau}{m_e}$$

Drude Model

Question: A copper wire has an electron number density of $n = 8.5 \times 10^{28}/m^3$, The relaxation time for conduction electrons is $\tau = 2.5 \times 10^{-14} s$. An electric field of $E = 0.02 V/m$ is applied across the wire. Using the Drude model, calculate the following:

- Drift velocity of the electrons.
- Electrical conductivity of copper.
- Current density in the wire.
- Mean free path of the electrons, assuming their average thermal velocity is $v_{th} = 1.0 \times 10^6 m/s$

$$\begin{aligned} v_{dx} &= \frac{eE_x\tau}{m_e} \\ &= \frac{1.6 * 10^{-19} * 0.02 * 2.5 * 10^{-14}}{9.1 * 10^{-31}} m/s \\ &= \frac{1.6 * 10^{-19} * 0.02 * 2.5 * 10^{-14}}{9.1 * 10^{-31}} m/s \\ &= 8.79 * 10^{-5} m/s \end{aligned}$$

$$\begin{aligned} \sigma &= \frac{e^2 n \tau}{m_e} \\ &= \frac{(1.6 * 10^{-19})^2 * 8.5 * 10^{28} * 2.5 * 10^{-14}}{9.1 * 10^{-31}} S/m \\ &= 5.97 * 10^7 S/m \\ J_x &= \sigma E_x = 1.20 * 10^6 A/m^2 \\ \lambda &= v_{th} \tau = 2.5 * 10^{-8} m = 25 nm \end{aligned}$$

Hall Effect

- Hall effect is a consequences of the forces that are exerted on moving charges by electric and magnetic fields.
- Hall Effect is used to distinguish whether a semiconductor is *n – type* or *p – type*.
- It is also used to measure the majority carrier concentration and majority carrier mobility.

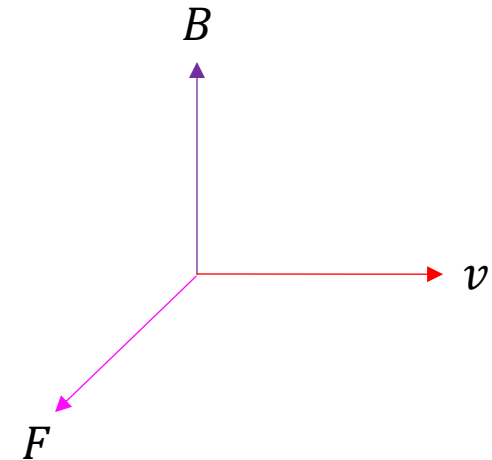
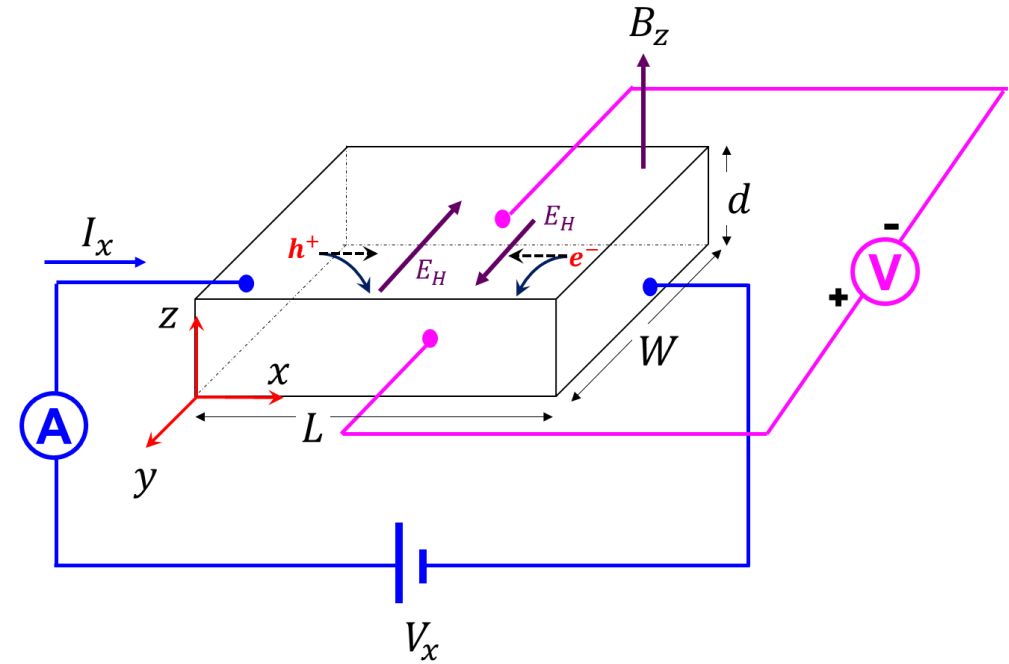
The force on a particle having charge q and moving in a magnetic field B is given by,

$$F = qv \times B$$

The force is perpendicular to both velocity and magnetic field.

Hall Effect

- A semiconductor with current I_x is placed in a magnetic field (B_z) perpendicular to the current.
- Due to the presence of the magnetic field (B_z), the carriers (either electrons or holes) will experience a force in the (negative or positive) y direction.
- This will lead to a charge build-up and an electric field in the y- direction.
- In steady state, the magnetic field will be balanced by the induced electric field.



Hall Effect

- The induced electric field in the y-direction is called Hall field. The Hall field produces a voltage across the semiconductor which is called the Hall Voltage.

$$V_H = E_H W$$

Now, If the carriers are holes

$$qE_y + q(v_x \times B_z) = 0$$

$$E_H = -E_y$$

$$qE_y = -q(v_x \times B_z)$$

$$E_H = v_x B_z$$

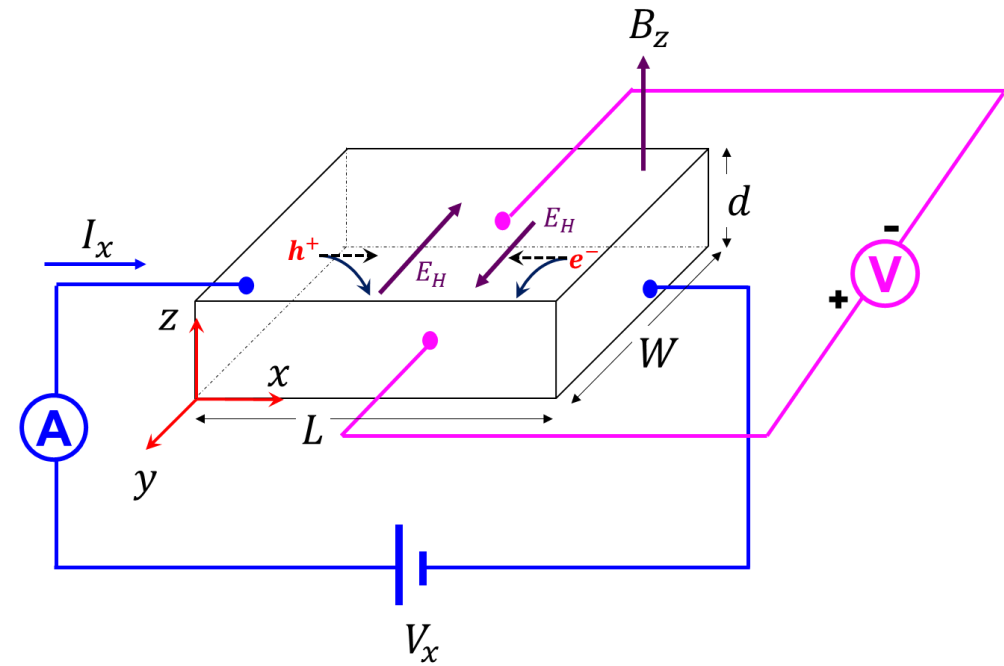
$$E_y = -v_x B_z$$

$$V_H = v_x B_z W$$

The drift velocity of holes can be written as

$$v_x = \frac{J_x}{ep} = \frac{I_x}{epWd}$$

$$V_H = \frac{I_x}{epWd} B_z W = \frac{I_x B_z}{epd}$$



$$V_H = \frac{I_x B_z}{epd}$$

$$p = \frac{I_x B_z}{edV_H}$$

Hall Effect

Now, If the carriers are electrons

$$V_H = -E_H W$$

$$-qE_y - q(v_x \times B_z) = 0$$

$$E_H = E_y$$

$$-qE_y = q(v_x \times B_z)$$

$$E_H = -v_x B_z$$

$$E_y = -v_x B_z$$

$$V_H = v_x B_z W$$

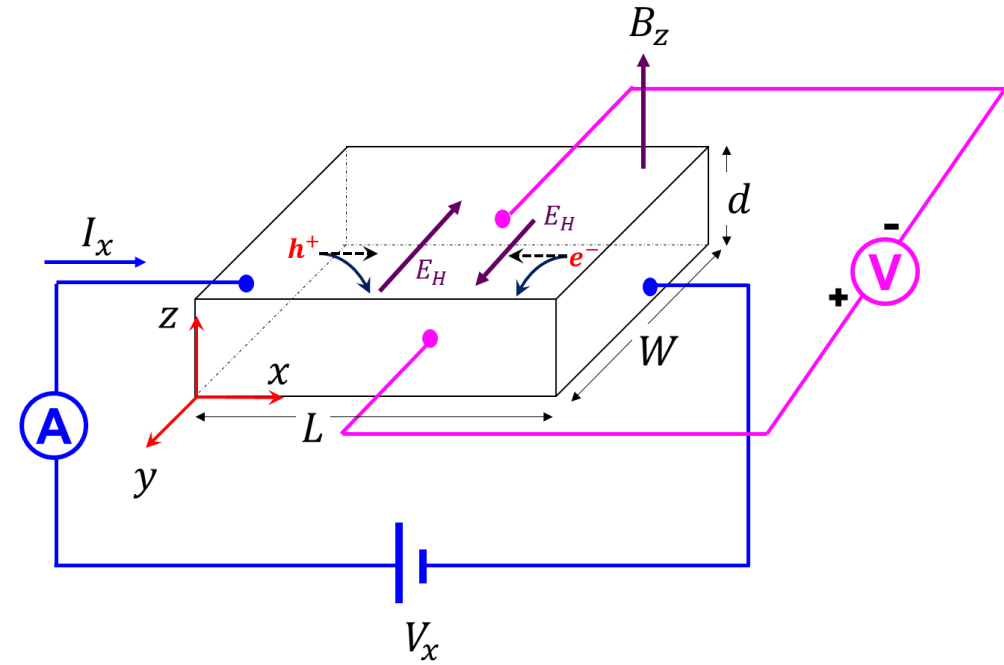
The drift velocity of holes can be written as

$$v_x = \frac{J_x}{en} = \frac{I_x}{enWd}$$

$$V_H = \frac{I_x}{enWd} B_z W = \frac{I_x B_z}{end}$$

$$V_H = \frac{I_x B_z}{end}$$

$$n = \frac{I_x B_z}{edV_H}$$



Hole mobility

$$v_x = \frac{J_x}{ep}$$

$$\mu_p E_x = \frac{J_x}{ep}$$

$$\frac{\mu_p V_x}{L} = \frac{J_x}{ep}$$

$$\mu_p = \frac{J_x L}{ep V_x}$$

$$\mu_p = \frac{I_x L}{ep V_x W d}$$

Electron mobility

$$v_x = \frac{J_x}{en}$$

$$\mu_n(E_x) = \frac{J_x}{en}$$

$$\frac{\mu_n V_x}{L} = \frac{J_x}{en}$$

$$\mu_n = \frac{J_x L}{en V_x}$$

$$\mu_n = \frac{I_x L}{en V_x W d}$$

Hall Coefficient

$$V_H = \frac{I_x B_z}{epd} = \frac{J_x B_z}{ep} = R_H J_x B_z$$

$$V_H = \frac{I_x B_z}{end} = \frac{J_x B_z}{en} = R_H J_x B_z$$

R_H is called Hall coefficient

$$R_H = \frac{p\mu_p^2 - n\mu_n^2}{q(p\mu_p + n\mu_n)^2}$$

R_H for p-type $p \gg n$

$$R_H = \frac{p\mu_p^2}{q(p\mu_p)^2} = \frac{1}{qp}$$

R_H for n-type $n \gg p$

$$R_H = \frac{-n\mu_n^2}{q(n\mu_n)^2} = \frac{1}{-qn}$$